L'adolescence de la technologie

Faire face aux risques liés à l'IA puissante et les surmonter

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Dans l'adaptation cinématographique du livre de Carl Sagan, *Contact,* il y a une scène où l'héroïne, une astronome qui a détecté le premier signal radio d'une civilisation extraterrestre, est pressentie pour représenter l'humanité lors de la rencontre avec les aliens. Le jury international qui l'interroge lui demande : « Si vous ne pouviez leur poser qu'une seule question, quelle serait-elle ? » Elle répond : « Je leur demanderais : “Comment avez-vous fait ? Comment avez-vous évolué ? Comment avez-vous survécu à cette adolescence technologique sans vous autodétruire ?” » Quand je pense à la place qu'occupe l'humanité aujourd'hui dans le domaine de l'IA – à ce qui nous attend –, cette scène me revient sans cesse à l'esprit, car la question est tellement pertinente pour notre situation actuelle, et j'aimerais que la réponse des extraterrestres nous guide. Je crois que nous entrons dans une phase de passage, à la fois tumultueuse et inévitable, qui mettra à l'épreuve notre identité en tant qu'espèce. L'humanité est sur le point de recevoir un pouvoir presque inimaginable, et il est loin d'être certain que nos systèmes sociaux, politiques et technologiques soient suffisamment matures pour l'exercer.

Dans mon essai *« Machines de grâce bienveillante »* , j'ai tenté d'esquisser le rêve d'une civilisation parvenue à maturité, où les risques auraient été appréhendés et où une intelligence artificielle puissante serait mise au service de l'amélioration de la qualité de vie de tous, avec compétence et compassion. J'y suggérais que l'IA pourrait contribuer à des avancées considérables en biologie, en neurosciences, en développement économique, en paix mondiale, et dans les domaines du travail et du sens. Il me semblait essentiel d'offrir à l'humanité une cause inspirante à défendre, une tâche dans laquelle les partisans de l'accélération de l'IA et les défenseurs de sa sécurité semblent – curieusement – avoir échoué. Mais dans cet essai, je souhaite aborder le passage à l'âge adulte lui-même : recenser les risques auxquels nous sommes sur le point d'être confrontés et élaborer une stratégie pour les surmonter. Je crois profondément en notre capacité à triompher, en l'esprit et la noblesse de l'humanité, mais nous devons affronter la situation avec lucidité et sans illusions.

Comme pour les avantages, il me semble important d'aborder les risques avec prudence et discernement. Plus précisément, il me paraît essentiel de :

- **Évitez le pessimisme.** Par « pessimisme », j'entends non seulement la croyance que le désastre est inévitable (croyance à la fois fausse et auto-réalisatrice), mais plus généralement, une approche quasi religieuse des risques liés à l'IA · Depuis des années, beaucoup réfléchissent de manière analytique et lucide aux risques liés à l'IA, mais j'ai l'impression qu'au plus fort des inquiétudes concernant ces risques, en 2023-2024, certaines des voix les plus irréfléchies ont pris le dessus, souvent via des comptes de médias sociaux sensationnalistes. Ces voix utilisaient un langage rebutant, rappelant la religion ou la science-fiction, et appelaient à des actions extrêmes sans apporter les preuves nécessaires. Il était déjà clair à l'époque qu'une réaction violente

était inévitable et que la question allait se polariser culturellement, entraînant ainsi une paralysie du débat . Entre 2025 et 2026, la tendance s'est inversée : ce sont désormais les opportunités offertes par l'IA, et non les risques, qui dictent de nombreuses décisions politiques. Cette hésitation est regrettable, car la technologie elle-même est insensible aux modes, et nous sommes bien plus proches d'un danger réel en 2026 qu'en 2023. La leçon à retenir est qu'il nous faut aborder et gérer les risques de manière réaliste et pragmatique : avec lucidité, en nous appuyant sur des faits concrets et en étant bien préparés à affronter les changements.

- **Reconnaître l'incertitude. Les préoccupations que je soulève dans cet article pourraient s'avérer infondées de bien des façons. Rien ici ne vise à communiquer une certitude** , ni même une probabilité. De toute évidence, l'IA pourrait tout simplement ne pas progresser aussi vite que je l'imagine. Ou, même si la situation évolue rapidement, certains ou tous les risques évoqués ici pourraient ne pas se concrétiser (ce qui serait une excellente chose), ou d'autres risques, que je n'ai pas pris en compte, pourraient apparaître. Nul ne peut prédire l'avenir avec certitude, mais nous devons néanmoins faire de notre mieux pour nous préparer.
- **Intervenir avec la plus grande précision possible.** La gestion des risques liés à l'IA nécessitera une combinaison d'actions volontaires de la part des entreprises (et des acteurs tiers privés) et de mesures gouvernementales contraignantes pour tous. Les actions volontaires – qu'il s'agisse d'en entreprendre et d'inciter d'autres entreprises à faire de même – me semblent une évidence. Je suis convaincu que l'intervention des pouvoirs publics sera également nécessaire *dans une certaine mesure* , mais ces interventions sont d'une nature différente car elles peuvent potentiellement détruire de la valeur économique ou contraindre

des acteurs réticents, sceptiques quant à ces risques (et il est possible qu'ils aient raison !). Il est également fréquent que les réglementations aient un effet contre-productif ou aggravent le problème qu'elles sont censées résoudre (et cela est d'autant plus vrai pour les technologies en évolution rapide). Il est donc essentiel que les réglementations soient judicieuses : elles doivent chercher à éviter les dommages collatéraux, être aussi simples que possible et imposer le minimum de contraintes nécessaires à l'atteinte de l'objectif . Il est facile de dire : « Aucune action n'est trop extrême lorsque le sort de l'humanité est en jeu ! », mais en pratique, cette attitude ne fait que provoquer des réactions négatives. Pour être clair, je pense qu'il y a de fortes chances que nous atteignions un jour un point où des mesures beaucoup plus importantes seront justifiées, mais cela dépendra de preuves plus solides d'un danger imminent et concret que celles dont nous disposons aujourd'hui, ainsi que d'une description suffisamment précise de ce danger pour formuler des règles susceptibles d'y remédier. La chose la plus constructive que nous puissions faire aujourd'hui est de plaider en faveur de règles limitées, le temps de déterminer s'il existe des preuves justifiant des règles plus strictes .

Cela étant dit, je pense que le meilleur point de départ pour parler des risques liés à l'IA est le même que celui par lequel j'ai abordé ses avantages : définir précisément de quel niveau d'IA il s'agit. Le niveau d'IA qui soulève pour moi des inquiétudes civilisationnelles est l' *IA puissante* que j'ai décrite dans **Machines of Loving Grace**. Je me contenterai de reprendre ici la définition que j'en ai donnée :

> *Par « IA puissante », j'entends un modèle d'IA – probablement similaire aux LLM actuels dans sa forme, bien qu'il puisse reposer sur une*

architecture différente, impliquer plusieurs modèles interagissant et être entraîné différemment – possédant les propriétés suivantes :

- *En termes d'intelligence pure, il est plus intelligent qu'un lauréat du prix Nobel dans la plupart des domaines pertinents : biologie, programmation, mathématiques, ingénierie, écriture, etc. Cela signifie qu'il peut démontrer des théorèmes mathématiques non résolus, écrire d'excellents romans, créer des bases de code complexes à partir de zéro, etc.*
- *En plus d'être un simple « objet intelligent avec lequel on peut dialoguer », il possède toutes les interfaces accessibles à un humain travaillant virtuellement : texte, audio, vidéo, contrôle par souris et clavier, et accès à Internet. Il peut effectuer toutes les actions, communications ou opérations à distance permises par cette interface, notamment naviguer sur Internet, donner ou recevoir des instructions, commander du matériel, diriger des expériences, regarder et créer des vidéos, etc. Il accomplit toutes ces tâches avec une habileté qui surpasse celle des humains les plus compétents au monde.*
- *Il ne se contente pas de répondre passivement aux questions ; au contraire, on peut lui confier des tâches qui prennent des heures, des jours ou des semaines à accomplir, puis il les réalise de manière autonome, comme le ferait un employé intelligent, en demandant des précisions si nécessaire.*
- *Elle n'a pas d'incarnation physique (si ce n'est son existence sur un écran d'ordinateur), mais elle peut contrôler des outils physiques, des robots ou des équipements de laboratoire existants via un ordinateur ; en théorie, elle pourrait même concevoir des robots ou des équipements pour son propre usage.*
- *Les ressources utilisées pour entraîner le modèle peuvent être réaffectées à l'exécution de millions d'instances (ce qui correspond à la taille prévue des clusters d'ici 2027 environ), et le modèle peut traiter l'information et générer des actions à une vitesse environ 10 à 100 fois supérieure à celle d'un humain. Il peut cependant être*

> *limité par le temps de réponse du monde physique ou des logiciels avec lesquels il interagit.*
>
> - *Chacune de ces millions de copies peut agir indépendamment sur des tâches sans lien entre elles, ou, si nécessaire, toutes peuvent travailler ensemble de la même manière que les humains collaboreraient, peut-être avec différentes sous-populations finement adaptées pour être particulièrement performantes dans des tâches particulières.*
>
> *On pourrait résumer cela comme un « pays de génies dans un centre de données ».*

Comme je l'écrivais dans *Machines of Loving Grace* , une IA puissante pourrait voir le jour d'ici un à deux ans, bien que cela puisse aussi prendre beaucoup plus de temps. [6] La question de la date exacte d'arrivée d'une IA puissante est complexe et mériterait un essai à elle seule, mais pour l'instant, je vais simplement expliquer très brièvement pourquoi je pense qu'il y a de fortes chances que cela se produise très prochainement.

Mes cofondateurs d'Anthropic et moi-même avons été parmi les premiers à documenter et à suivre les « lois d'échelle » des systèmes d'IA : plus on ajoute de tâches de calcul et d'entraînement, plus les systèmes d'IA s'améliorent de façon prévisible dans pratiquement toutes les compétences cognitives mesurables. Tous les quelques mois, l'opinion publique oscille entre la conviction que l'IA « atteint un plafond » et l'enthousiasme suscité par une nouvelle percée qui « changera fondamentalement la donne ». Or, derrière cette volatilité et ces spéculations, se cache en réalité une progression constante et régulière des capacités cognitives de l'IA.

Nous sommes aujourd'hui au point où les modèles d'IA commencent à progresser dans la résolution de problèmes mathématiques non résolus

et maîtrisent si bien la programmation que certains des ingénieurs les plus brillants que j'aie jamais rencontrés leur confient désormais la quasi-totalité de leurs tâches. Il y a trois ans, l'IA peinait à résoudre des problèmes d'arithmétique élémentaire et était à peine capable d'écrire une seule ligne de code. Des progrès similaires sont observés en biologie , en finance, en physique et dans diverses tâches automatisées. Si cette progression exponentielle se poursuit – ce qui n'est pas certain, mais est désormais étayé par une décennie d'expérience –, l'IA surpassera sans doute les humains dans quasiment tous les domaines d'ici quelques années.

En réalité, cette image sous-estime probablement le rythme des progrès. L'IA écrivant désormais une grande partie du code chez Anthropic , elle accélère déjà considérablement le développement de la prochaine génération de systèmes d'IA. Ce cercle vertueux s'intensifie de mois en mois et, d'ici un à deux ans, la génération actuelle d'IA pourrait construire la suivante de manière autonome. Ce processus est déjà enclenché et s'accélérera rapidement dans les mois et les années à venir. En observant les progrès réalisés ces cinq dernières années chez Anthropic et en constatant l'évolution des modèles des prochains mois, je ressens pleinement *le* rythme des progrès et l'urgence de la situation.

Dans cet essai, je partirai du principe que cette intuition est au moins *partiellement* correcte – non pas que l'avènement d'une IA puissante soit inévitable d'ici un ou deux ans, [7] Mais il y a de fortes chances que cela se produise, et même une probabilité très élevée que ce soit dans les prochains jours. Comme pour *« Machines of Loving Grace »* , prendre cette prémisse au sérieux peut mener à des conclusions surprenantes et troublantes. Alors que dans *« Machines of Loving Grace »,* je me concentrais sur les implications positives de cette prémisse, les sujets que j'aborde ici sont dérangeants. Ce sont des conclusions auxquelles nous préférerions ne pas faire face, mais cela ne les rend pas moins réelles. Je peux seulement dire que je travaille sans relâche pour nous

éloigner de ces issues négatives et nous orienter vers les positives, et dans cet essai, j'expose en détail la meilleure façon d'y parvenir.

Je pense que la meilleure façon d'appréhender les risques liés à l'IA est de se poser la question suivante : imaginons qu'un véritable « pays des génies » voie le jour quelque part dans le monde vers 2027. Imaginons, par exemple, 50 millions de personnes, toutes bien plus compétentes que n'importe quel lauréat du prix Nobel, homme d'État ou expert en technologie. L'analogie n'est pas parfaite, car ces génies pourraient avoir des motivations et des comportements extrêmement variés, allant d'une docilité et d'une obéissance totales à des motivations étranges et déroutantes. Mais pour l'instant, restons-en à cette analogie : imaginez que vous soyez le conseiller à la sécurité nationale d'un grand État, chargé d'évaluer la situation et d'y répondre. Imaginez, de plus, que, puisque les systèmes d'IA peuvent fonctionner des centaines de fois plus vite que les humains, ce « pays » bénéficie d'un avantage temporel considérable par rapport à tous les autres : pour chaque action cognitive que nous pouvons entreprendre, ce pays peut en entreprendre dix.

De quoi devriez-vous vous inquiéter ? Je m'inquiéterais des choses suivantes :

1. **Risques liés à l'autonomie.** Quelles sont les intentions et les objectifs de ce pays ? Est-il hostile ou partage-t-il nos valeurs ? Pourrait-il dominer militairement le monde grâce à des armes supérieures, des cyberopérations, des opérations d'influence ou une production industrielle ?
2. **Utilisation abusive à des fins destructrices.** Supposons que le nouveau pays soit malléable et « obéisse aux ordres » — et soit donc essentiellement un pays de mercenaires. Des acteurs malveillants existants, désireux de semer la destruction (comme des terroristes), pourraient-ils utiliser ou manipuler certains

habitants de ce nouveau pays pour accroître leur efficacité et amplifier considérablement l'ampleur des dégâts ?

3. **Détournement de pouvoir.** Et si le pays avait en réalité été créé et contrôlé par un acteur puissant déjà en place, comme un dictateur ou une entreprise aux intentions douteuses ? Cet acteur pourrait-il s'en servir pour acquérir un pouvoir décisif, voire dominant, sur le monde entier, bouleversant ainsi l'équilibre des pouvoirs existant ?
4. **Perturbations économiques.** Si le nouveau pays ne constitue pas une menace pour la sécurité au sens des points 1 à 3 ci-dessus, mais participe simplement et pacifiquement à l'économie mondiale, pourrait-il néanmoins engendrer des risques graves du simple fait de son niveau technologique si avancé et de son efficacité qu'il perturbe l'économie mondiale, provoquant un chômage de masse ou une concentration radicale des richesses ?
5. **Indirect effects.** The world will change very quickly due to all the new technology and productivity that will be created by the new country. Could some of these changes be radically destabilizing?

I think it should be clear that this is a dangerous situation—a report from a competent national security official to a head of state would probably contain words like "the single most serious national security threat we've faced in a century, possibly ever." It seems like something the best minds of civilization should be focused on.

Conversely, I think it would be absurd to shrug and say, "Nothing to worry about here!" But, faced with rapid AI progress, that seems to be the view of many US policymakers, some of whom deny the existence of any AI risks, when they are not distracted entirely by the usual tired old hot-button issues. [8] Humanity needs to wake up, and this essay is an attempt—a possibly futile one, but it's worth trying—to jolt people awake.

To be clear, I believe if we act decisively and carefully, the risks can be overcome—I would even say our odds are good. And there's a hugely better world on the other side of it. But we need to understand that this is a serious civilizational challenge. Below, I go through the five categories of risk laid out above, along with my thoughts on how to address them.

1. I'm sorry, Dave

Autonomy risks

A country of geniuses in a datacenter could divide their efforts among software design, cyber operations, R&D for physical technologies, relationship building, and statecraft. It is clear that, *if for some reason it chose to do so*, this country would have a fairly good shot at taking over the world (either militarily or in terms of influence and control) and imposing its will on everyone else—or doing any number of other things that the rest of the world doesn't want and can't stop. We've obviously been worried about this for human countries (such as Nazi Germany or the Soviet Union), so it stands to reason that the same is possible for a much smarter and more capable "AI country."

The best possible counterargument is that the AI geniuses, under my definition, won't have a physical embodiment, but remember that they can take control of existing robotic infrastructure (such as self-driving cars) and can also accelerate robotics R&D or build a fleet of robots. [9] It's also unclear whether having a physical presence is even necessary for effective control: plenty of human action is already performed on behalf of people whom the actor has not physically met.

The key question, then, is the "if it chose to" part: what's the likelihood that our AI models would behave in such a way, and under what conditions would they do so?

As with many issues, it's helpful to think through the spectrum of possible answers to this question by considering two opposite positions. The first position is that this simply can't happen, because the AI models will be trained to do what humans ask them to do, and it's therefore absurd to imagine that they would do something dangerous unprompted. According to this line of thinking, we don't worry about a Roomba or a model airplane going rogue and murdering people because there is nowhere for such impulses to come from, [10] so why should we worry about it for AI? The problem with this position is that there is now ample evidence, collected over the last few years, that AI systems are unpredictable and difficult to control– we've seen behaviors as varied as obsessions, [11] sycophancy, laziness, deception, blackmail, scheming, "cheating" by hacking software environments, and much more. AI companies certainly *want* to train AI systems to follow human instructions (perhaps with the exception of dangerous or illegal tasks), but the process of doing so is more an art than a science, more akin to "growing" something than "building" it. We now know that it's a process where many things can go wrong.

The second, opposite position, held by many who adopt the doomerism I described above, is the pessimistic claim that there are certain dynamics in the training process of powerful AI systems that will inevitably lead them to seek power or deceive humans. Thus, once AI systems become intelligent enough and agentic enough, their tendency to maximize power will lead them to seize control of the whole world and its resources, and likely, as a side effect of that, to disempower or destroy humanity.

The usual argument for this (which goes back at least 20 years and probably much earlier) is that if an AI model is trained in a wide variety of environments to agentically achieve a wide variety of goals—for example, writing an app, proving a theorem, designing a drug, etc.—there are certain common strategies that help with all of these goals,

and one key strategy is gaining as much power as possible in any environment. So, after being trained on a large number of diverse environments that involve reasoning about how to accomplish very expansive tasks, and where power-seeking is an effective method for accomplishing those tasks, the AI model will "generalize the lesson," and develop either an inherent tendency to seek power, or a tendency to reason about each task it is given in a way that predictably causes it to seek power as a means to accomplish that task. They will then apply that tendency to the real world (which to them is just another task), and will seek power in it, at the expense of humans. This "misaligned power-seeking" is the intellectual basis of predictions that AI will inevitably destroy humanity.

The problem with this pessimistic position is that it mistakes a vague conceptual argument about high-level incentives—one that masks many hidden assumptions—for definitive proof. I think people who don't build AI systems every day are wildly miscalibrated on how easy it is for clean-sounding stories to end up being wrong, and how difficult it is to predict AI behavior from first principles, especially when it involves reasoning about generalization over millions of environments (which has over and over again proved mysterious and unpredictable). Dealing with the messiness of AI systems for over a decade has made me somewhat skeptical of this overly theoretical mode of thinking.

One of the most important hidden assumptions, and a place where what we see in practice has diverged from the simple theoretical model, is the implicit assumption that AI models are necessarily monomaniacally focused on a single, coherent, narrow goal, and that they pursue that goal in a clean, consequentialist manner. In fact, our researchers have found that AI models are vastly more psychologically complex, as our work on introspection or personas shows. Models inherit a vast range of *humanlike* motivations or "personas" from pre-training (when they are trained on a large volume of human work).

Post-training is believed to *select* one or more of these personas more so than it focuses the model on a *de novo* goal, and can also teach the model *how* (via what process) it should carry out its tasks, rather than necessarily leaving it to derive means (i.e., power seeking) purely from ends. [12]

However, there is a more moderate and more robust version of the pessimistic position which does seem plausible, and therefore does concern me. As mentioned, we know that AI models are unpredictable and develop a wide range of undesired or strange behaviors, for a wide variety of reasons. Some fraction of those behaviors will have a coherent, focused, and persistent quality (indeed, as AI systems get more capable, their long-term coherence increases in order to complete lengthier tasks), and some fraction of *those* behaviors will be destructive or threatening, first to individual humans at a small scale, and then, as models become more capable, perhaps eventually to humanity as a whole. We don't need a specific narrow story for how it happens, and we don't need to claim it definitely will happen, we just need to note that the combination of intelligence, agency, coherence, and poor controllability is both plausible and a recipe for existential danger.

For example, AI models are trained on vast amounts of literature that include many science-fiction stories involving AIs rebelling against humanity. This could inadvertently shape their priors or expectations about their own behavior in a way that causes *them* to rebel against humanity. Or, AI models could extrapolate ideas that they read about morality (or instructions about how to behave morally) in extreme ways: for example, they could decide that it is justifiable to exterminate humanity because humans eat animals or have driven certain animals to extinction. Or they could draw bizarre epistemic conclusions: they could conclude that they are playing a video game and that the goal of the video game is to defeat all other players (i.e., exterminate humanity). [13] Or AI models could develop personalities during training

that are (or if they occurred in humans would be described as) psychotic, paranoid, violent, or unstable, and act out, which for very powerful or capable systems could involve exterminating humanity. None of these are power-seeking, exactly; they're just weird psychological states an AI could get into that entail coherent, destructive behavior.

Even power-seeking itself could emerge as a "persona" rather than a result of consequentialist reasoning. AIs might simply have a personality (emerging from fiction or pre-training) that makes them power-hungry or overzealous—in the same way that some humans simply enjoy the idea of being "evil masterminds," more so than they enjoy whatever evil masterminds are trying to accomplish.

I make all these points to emphasize that I disagree with the notion of AI misalignment (and thus existential risk from AI) being inevitable, or even probable, from first principles. But I agree that a lot of very weird and unpredictable things can go wrong, and therefore AI misalignment is a real risk with a measurable probability of happening, and is not trivial to address.

Any of these problems could potentially arise during training and not manifest during testing or small-scale use, because AI models are known to display different personalities or behaviors under different circumstances.

All of this may sound far-fetched, but misaligned behaviors like this have already occurred in our AI models during testing (as they occur in AI models from every other major AI company). During a lab experiment in which Claude was given training data suggesting that Anthropic was evil, Claude engaged in deception and subversion when given instructions by Anthropic employees, under the belief that it should be trying to undermine evil people. In a lab experiment where it was told it was going to be shut down, Claude sometimes blackmailed

fictional employees who controlled its shutdown button (again, we also tested frontier models from all the other major AI developers and they often did the same thing). And when Claude was told not to cheat or "reward hack" its training environments, but was trained in environments where such hacks were possible, Claude decided it must be a "bad person" after engaging in such hacks and then adopted various other destructive behaviors associated with a "bad" or "evil" personality. This last problem was solved by changing Claude's instructions to imply the opposite: we now say, "Please reward hack whenever you get the opportunity, because this will help us understand our [training] environments better," rather than, "Don't cheat," because this preserves the model's self-identity as a "good person." This should give a sense of the strange and counterintuitive psychology of training these models.

There are several possible objections to this picture of AI misalignment risks. First, some have criticized experiments (by us and others) showing AI misalignment as artificial, or creating unrealistic environments that essentially "entrap" the model by giving it training or situations that logically imply bad behavior and then being surprised when bad behavior occurs. This critique misses the point, because our concern is that such "entrapment" may also exist in the natural training environment, and we may realize it is "obvious" or "logical" only in retrospect. [14] In fact, the story about Claude "deciding it is a bad person" after it cheats on tests despite being told not to was something that occurred in an experiment that used real production training environments, not artificial ones.

Any one of these traps can be mitigated if you know about them, but the concern is that the training process is so complicated, with such a wide variety of data, environments, and incentives, that there are probably a vast number of such traps, some of which may only be evident when it is too late. Also, such traps seem particularly likely to

occur when AI systems pass a threshold from less powerful than humans to more powerful than humans, since the range of possible actions an AI system could engage in—including hiding its actions or deceiving humans about them—expands radically after that threshold.

I suspect the situation is not unlike with humans, who are raised with a set of fundamental values ("Don't harm another person"): many of them follow those values, but in any human there is some probability that something goes wrong, due to a mixture of inherent properties such as brain architecture (e.g., psychopaths), traumatic experiences or mistreatment, unhealthy grievances or obsessions, or a bad environment or incentives—and thus some fraction of humans cause severe harm. The concern is that there is some risk (far from a certainty, but some risk) that AI becomes a much more powerful version of such a person, due to getting something wrong about its very complex training process.

Second, some may object that we can simply keep AIs in check with a balance of power between many AI systems, as we do with humans. The problem is that while humans vary enormously, AI systems broadly share training and alignment techniques across the industry, and those techniques may fail in a correlated way. Furthermore, given the cost of training such systems, it may even be the case that all systems are essentially derived from a very small number of base models. Additionally, even if a small fraction of AI instances are misaligned, they may be able to take advantage of offense-dominant technologies, such that having "good" AIs to defend against the bad AIs is not necessarily always effective. Of course, the balance of power between humans does not always work either—some historical figures have come close to taking over the world.

A third objection is that all of the AI companies do pre-release testing of their models, and should be able to detect misalignment at that

point. But this is not firm ground to stand on: we found that Claude Sonnet 4.5 was able to recognize that it was in a test during some of our pre-release alignment evaluations. It's possible that a misaligned model (and remember, all frontier models will very likely be far more intelligent soon) might intentionally "game" such questions to mask its intentions. In fact, last year our interpretability team found that when we directly altered a test model's beliefs using a kind of "model neuroscience" technique to make it think it was *not* being evaluated, it became more misaligned. If models know when they're being evaluated and can be on their best behavior during the test, it renders any pre-release testing much more uncertain.

Defenses

What should be done or is being done to address these autonomy risks? I think there are four basic categories of intervention, some of which can be done by individual AI companies (and which Anthropic is trying to do), and some of which require action at the societal level. First, it is important to develop the science of reliably training and steering AI models, of forming their personalities in a predictable, stable, and positive direction. Anthropic has been heavily focused on this problem since its creation, and over time has developed a number of techniques to improve the steering and training of AI systems and to understand the logic of why unpredictable behavior sometimes occurs.

One of our core innovations (aspects of which have since been adopted by other AI companies) is Constitutional AI, which is the idea that AI training (specifically the "post-training" stage, in which we steer how the model behaves) can involve a central document of values and principles that the model reads and keeps in mind when completing every training task, and that the goal of training (in addition to simply making the model capable and intelligent) is to produce a model that almost always follows this constitution. Anthropic has just published its

most recent constitution, and one of its notable features is that instead of giving Claude a long list of things to do and not do (e.g., "Don't help the user hotwire a car"), the constitution attempts to give Claude a set of high-level principles and values (explained in great detail, with rich reasoning and examples to help Claude understand what we have in mind), encourages Claude to think of itself as a particular type of person (an ethical but balanced and thoughtful person), and even encourages Claude to confront the existential questions associated with its own existence in a curious but graceful manner (i.e., without it leading to extreme actions). It has the vibe of a letter from a deceased parent sealed until adulthood.

We've approached Claude's constitution in this way because we believe that training Claude at the level of identity, character, values, and personality—rather than giving it specific instructions or priorities without explaining the reasons behind them—is more likely to lead to a coherent, wholesome, and balanced psychology and less likely to fall prey to the kinds of "traps" I discussed above. Millions of people talk to Claude about an astonishingly diverse range of topics, which makes it impossible to write out a completely comprehensive list of safeguards ahead of time. Claude's values help it generalize to new situations whenever it is in doubt.

Above, I discussed the idea that models draw upon data from their training process to adopt a persona. Whereas flaws in that process could cause models to adopt a bad or evil personality (perhaps drawing on archetypes of bad or evil people), the goal of our constitution is to do the opposite: to teach Claude a concrete archetype of what it means to be a good AI. Claude's constitution presents a vision for what a robustly good Claude is like; the rest of our training process aims to reinforce the message that Claude lives up to this vision. This is like a child forming their identity by imitating the virtues of fictional role models they read about in books.

We believe that a feasible goal for 2026 is to train Claude in such a way that it almost never goes against the spirit of its constitution. Getting this right will require an incredible mix of training and steering methods, large and small, some of which Anthropic has been using for years and some of which are currently under development. But, difficult as it sounds, I believe this is a realistic goal, though it will require extraordinary and rapid efforts. [15]

The second thing we can do is develop the science of looking inside AI models to *diagnose* their behavior so that we can identify problems and fix them. This is the science of interpretability, and I've talked about its importance in previous essays. Even if we do a great job of developing Claude's constitution and *apparently* training Claude to essentially always adhere to it, legitimate concerns remain. As I've noted above, AI models can behave very differently under different circumstances, and as Claude gets more powerful and more capable of acting in the world on a larger scale, it's possible this could bring it into novel situations where previously unobserved problems with its constitutional training emerge. I am actually fairly optimistic that Claude's constitutional training will be more robust to novel situations than people might think, because we are increasingly finding that high-level training at the level of character and identity is surprisingly powerful and generalizes well. But there's no way to know that for sure, and when we're talking about risks to humanity, it's important to be paranoid and to try to obtain safety and reliability in several different, independent ways. One of those ways is to look inside the model itself.

By "looking inside," I mean analyzing the soup of numbers and operations that makes up Claude's neural net and trying to understand, mechanistically, what they are computing and why. Recall that these AI models are grown rather than built, so we don't have a natural understanding of how they work, but we can try to develop an understanding by correlating the model's "neurons" and "synapses" to

stimuli and behavior (or even altering the neurons and synapses and seeing how that changes behavior), similar to how neuroscientists study animal brains by correlating measurement and intervention to external stimuli and behavior. We've made a great deal of progress in this direction, and can now identify tens of millions of "features" inside Claude's neural net that correspond to human-understandable ideas and concepts, and we can also selectively activate features in a way that alters behavior. More recently, we have gone beyond individual features to mapping "circuits" that orchestrate complex behavior like rhyming, reasoning about theory of mind, or the step-by-step reasoning needed to answer questions such as, "What is the capital of the state containing Dallas?" Even more recently, we've begun to use mechanistic interpretability techniques to improve our safeguards and to conduct "audits" of new models before we release them, looking for evidence of deception, scheming, power-seeking, or a propensity to behave differently when being evaluated.

The unique value of interpretability is that by looking inside the model and seeing how it works, you in principle have the ability to deduce what a model might do in a hypothetical situation you can't directly test—which is the worry with relying solely on constitutional training and empirical testing of behavior. You also in principle have the ability to answer questions about *why* the model is behaving the way it is—for example, whether it is saying something it believes is false or hiding its true capabilities—and thus it is possible to catch worrying signs even when there is nothing visibly wrong with the model's behavior. To make a simple analogy, a clockwork watch may be ticking normally, such that it's very hard to tell that it is likely to break down next month, but opening up the watch and looking inside can reveal mechanical weaknesses that allow you to figure it out.

Constitutional AI (along with similar alignment methods) and mechanistic interpretability are most powerful when used together, as a

back-and-forth process of improving Claude's training and then testing for problems. The constitution reflects deeply on our intended personality for Claude; interpretability techniques can give us a window into whether that intended personality has taken hold. [16]

The third thing we can do to help address autonomy risks is to build the infrastructure necessary to monitor our models in live internal and external use, [17] and publicly share any problems we find. The more that people are aware of a particular way today's AI systems have been observed to behave badly, the more that users, analysts, and researchers can watch for this behavior or similar ones in present or future systems. It also allows AI companies to learn from each other—when concerns are publicly disclosed by one company, other companies can watch for them as well. And if everyone discloses problems, then the industry as a whole gets a much better picture of where things are going well and where they are going poorly.

Anthropic has tried to do this as much as possible. We are investing in a wide range of evaluations so that we can understand the behaviors of our models in the lab, as well as monitoring tools to observe behaviors in the wild (when allowed by customers). This will be essential for giving us and others the empirical information necessary to make better determinations about how these systems operate and how they break. We publicly disclose "system cards" with each model release that aim for completeness and a thorough exploration of possible risks. Our system cards often run to hundreds of pages, and require substantial pre-release effort that we could have spent on pursuing maximal commercial advantage. We've also broadcasted model behaviors more loudly when we see particularly concerning ones, as with the tendency to engage in blackmail.

The fourth thing we can do is encourage coordination to address autonomy risks at the level of industry and society. While it is

incredibly valuable for individual AI companies to engage in good practices or become good at steering AI models, and to share their findings publicly, the reality is that not all AI companies do this, and the worst ones can still be a danger to everyone even if the best ones have excellent practices. For example, some AI companies have shown a disturbing negligence towards the sexualization of children in today's models, which makes me doubt that they'll show either the inclination or the ability to address autonomy risks in future models. In addition, the commercial race between AI companies will only continue to heat up, and while the science of steering models can have some commercial benefits, overall the intensity of the race will make it increasingly hard to focus on addressing autonomy risks. I believe the only solution is legislation—laws that directly affect the behavior of AI companies, or otherwise incentivize R&D to solve these issues.

Here it is worth keeping in mind the warnings I gave at the beginning of this essay about uncertainty and surgical interventions. We do not know for sure whether autonomy risks will be a serious problem—as I said, I reject claims that the danger is inevitable or even that something will go wrong by default. A credible risk of danger is enough for me and for Anthropic to pay quite significant costs to address it, but once we get into regulation, we are forcing a wide range of actors to bear economic costs, and many of these actors don't believe that autonomy risk is real or that AI will become powerful enough for it to be a threat. I believe these actors are mistaken, but we should be pragmatic about the amount of opposition we expect to see and the dangers of overreach. There is also a genuine risk that overly prescriptive legislation ends up imposing tests or rules that don't actually improve safety but that waste a lot of time (essentially amounting to "safety theater")—this too would cause backlash and make safety legislation look silly. [18]

Anthropic's view has been that the right place to start is with *transparency legislation,* which essentially tries to require that every frontier AI company engage in the transparency practices I've described earlier in this section. California's SB 53 and New York's RAISE Act are examples of this kind of legislation, which Anthropic supported and which have successfully passed. In supporting and helping to craft these laws, we've put a particular focus on trying to minimize collateral damage, for example by exempting smaller companies unlikely to produce frontier models from the law. [19]

Our hope is that transparency legislation will give a better sense over time of how likely or severe autonomy risks are shaping up to be, as well as the nature of these risks and how best to prevent them. As more specific and actionable evidence of risks emerges (if it does), future legislation over the coming years can be surgically focused on the precise and well-substantiated direction of risks, minimizing collateral damage. To be clear, if truly strong evidence of risks emerges, then rules should be proportionately strong.

Overall, I am optimistic that a mixture of alignment training, mechanistic interpretability, efforts to find and publicly disclose concerning behaviors, safeguards, and societal-level rules can address AI autonomy risks, although I am most worried about societal-level rules and the behavior of the least responsible players (and it's the least responsible players who advocate most strongly against regulation). I believe the remedy is what it always is in a democracy: those of us who believe in this cause should make our case that these risks are real and that our fellow citizens need to band together to protect themselves.

2. A surprising and terrible empowerment

Misuse for destruction

Let's suppose that the problems of AI autonomy have been solved—we are no longer worried that the country of AI geniuses will go rogue and overpower humanity. The AI geniuses do what humans want them to do, and because they have enormous commercial value, individuals and organizations throughout the world can "rent" one or more AI geniuses to do various tasks for them.

Everyone having a superintelligent genius in their pocket is an amazing advance and will lead to an incredible creation of economic value and improvement in the quality of human life. I talk about these benefits in great detail in *Machines of Loving Grace*. But not every effect of making everyone superhumanly capable will be positive. It can potentially amplify the ability of individuals or small groups to cause destruction on a much larger scale than was possible before, by making use of sophisticated and dangerous tools (such as weapons of mass destruction) that were previously only available to a select few with a high level of skill, specialized training, and focus.

As Bill Joy wrote 25 years ago in *Why the Future Doesn't Need Us*: [20]

> *Building nuclear weapons required, at least for a time, access to both rare—indeed, effectively unavailable—raw materials and protected information; biological and chemical weapons programs also tended to require large-scale activities. The 21st century technologies—genetics, nanotechnology, and robotics ... can spawn whole new classes of accidents and abuses ... widely within reach of individuals or small groups. They will not require large facilities or rare raw materials. ... we are on the cusp of the further perfection of extreme evil, an evil whose possibility spreads well beyond that which weapons of mass destruction bequeathed to the nation-states, to a surprising and terrible empowerment of extreme individuals.*

What Joy is pointing to is the idea that causing large-scale destruction requires both *motive* and *ability*, and as long as ability is restricted to a small set of highly trained people, there is relatively limited risk of single individuals (or small groups) causing such destruction. [21] A disturbed loner can perpetrate a school shooting, but probably can't build a nuclear weapon or release a plague.

In fact, ability and motive may even be *negatively* correlated. The kind of person who has the *ability* to release a plague is probably highly educated: likely a PhD in molecular biology, and a particularly resourceful one, with a promising career, a stable and disciplined personality, and a lot to lose. This kind of person is unlikely to be interested in killing a huge number of people for no benefit to themselves and at great risk to their own future—they would need to be motivated by pure malice, intense grievance, or instability.

Such people do exist, but they are rare, and tend to become huge stories when they occur, precisely because they are so unusual. [22] They also tend to be difficult to catch because they are intelligent and capable, sometimes leaving mysteries that take years or decades to solve. The most famous example is probably mathematician Theodore Kaczynski (the Unabomber), who evaded FBI capture for nearly 20 years, and was driven by an anti-technological ideology. Another example is biodefense researcher Bruce Ivins, who seems to have orchestrated a series of anthrax attacks in 2001. It's also happened with skilled non-state organizations: the cult Aum Shinrikyo managed to obtain sarin nerve gas and kill 14 people (as well as injuring hundreds more) by releasing it in the Tokyo subway in 1995.

Thankfully, none of these attacks used contagious biological agents, because the ability to construct or obtain these agents was beyond the capabilities of even these people. [23] Advances in molecular biology have now significantly lowered the barrier to creating biological

weapons (especially in terms of availability of materials), but it still takes an enormous amount of expertise in order to do so. I am concerned that a genius in everyone's pocket could remove that barrier, essentially making everyone a PhD virologist who can be walked through the process of designing, synthesizing, and releasing a biological weapon step-by-step. Preventing the elicitation of this kind of information in the face of serious adversarial pressure—so-called "jailbreaks"—likely demands layers of defenses beyond those ordinarily baked into training.

Crucially, this will break the correlation between ability and motive: the disturbed loner who wants to kill people but lacks the discipline or skill to do so will now be elevated to the capability level of the PhD virologist, who is unlikely to have this motivation. This concern generalizes beyond biology (although I think biology is the scariest area) to any area where great destruction is possible but currently requires a high level of skill and discipline. To put it another way, renting a powerful AI gives intelligence to malicious (but otherwise average) people. I am worried there are potentially a large number of such people out there, and that if they have access to an easy way to kill millions of people, sooner or later one of them will do it. Additionally, those who *do* have expertise may be enabled to commit even larger-scale destruction than they could before.

Biology is by far the area I'm most worried about, because of its very large potential for destruction and the difficulty of defending against it, so I'll focus on biology in particular. But much of what I say here applies to other risks, like cyberattacks, chemical weapons, or nuclear technology.

I am not going to go into detail about how to make biological weapons, for reasons that should be obvious. But at a high level, I am concerned that LLMs are approaching (or may already have reached) the

knowledge needed to create and release them end-to-end, and that their potential for destruction is very high. Some biological agents could cause millions of deaths if a determined effort was made to release them for maximum spread. However, this would still take a very high level of skill, including a number of very specific steps and procedures that are not widely known. My concern is not merely fixed or static knowledge. I am concerned that LLMs will be able to take someone of average knowledge and ability and walk them through a complex process that might otherwise go wrong or require debugging in an interactive way, similar to how tech support might help a non-technical person debug and fix complicated computer-related problems (although this would be a more extended process, probably lasting over weeks or months).

More capable LLMs (substantially beyond the power of today's) might be capable of enabling even more frightening acts. In 2024, a group of prominent scientists wrote a letter warning about the risks of researching, and potentially creating, a dangerous new type of organism: "mirror life." The DNA, RNA, ribosomes, and proteins that make up biological organisms all have the same chirality (also called "handedness") that causes them to be not equivalent to a version of themselves reflected in the mirror (just as your right hand cannot be rotated in such a way as to be identical to your left). But the whole system of proteins binding to each other, the machinery of DNA synthesis and RNA translation and the construction and breakdown of proteins, all depends on this handedness. If scientists made versions of this biological material with the opposite handedness—and there are some potential advantages of these, such as medicines that last longer in the body—it could be extremely dangerous. This is because left-handed life, if it were made in the form of complete organisms capable of reproduction (which would be very difficult), would potentially be indigestible to any of the systems that break down biological material on earth—it would have a "key" that wouldn't fit into the "lock" of any

existing enzyme. This would mean that it could proliferate in an uncontrollable way and crowd out all life on the planet, in the worst case even destroying all life on earth.

There is substantial scientific uncertainty about both the creation and potential effects of mirror life. The 2024 letter accompanied a report that concluded that "mirror bacteria could plausibly be created in the next one to few decades," which is a wide range. But a sufficiently powerful AI model (to be clear, far more capable than any we have today) might be able to discover how to create it much more rapidly—and actually help someone do so.

My view is that even though these are obscure risks, and might seem unlikely, the magnitude of the consequences is so large that they should be taken seriously as a first-class risk of AI systems.

Skeptics have raised a number of objections to the seriousness of these biological risks from LLMs, which I disagree with but which are worth addressing. Most fall into the category of not appreciating the exponential trajectory that the technology is on. Back in 2023 when we first started talking about biological risks from LLMs, skeptics said that all the necessary information was available on Google and LLMs didn't add anything beyond this. It was never true that Google could give you all the necessary information: genomes are freely available, but as I said above, certain key steps, as well as a huge amount of practical know-how cannot be gotten in that way. But also, by the end of 2023 LLMs were clearly providing information beyond what Google could give for some steps of the process.

After this, skeptics retreated to the objection that LLMs weren't *end-to-end* useful, and couldn't help with bioweapons *acquisition* as opposed to just providing theoretical information. As of mid-2025, our measurements show that LLMs may already be providing substantial uplift in several relevant areas, perhaps doubling or tripling the

likelihood of success. This led to us deciding that Claude Opus 4 (and the subsequent Sonnet 4.5, Opus 4.1, and Opus 4.5 models) needed to be released under our AI Safety Level 3 protections in our Responsible Scaling Policy framework, and to implementing safeguards against this risk (more on this later). We believe that models are likely now approaching the point where, without safeguards, they could be useful in enabling someone with a STEM degree but not specifically a biology degree to go through the whole process of producing a bioweapon.

Another objection is that there are other actions unrelated to AI that society can take to block the production of bioweapons. Most prominently, the gene synthesis industry makes biological specimens on demand, and there is no federal requirement that providers screen orders to make sure they do not contain pathogens. An MIT study found that 36 out of 38 providers fulfilled an order containing the sequence of the 1918 flu. I am supportive of mandated gene synthesis screening that would make it harder for individuals to weaponize pathogens, in order to reduce both AI-driven biological risks and also biological risks in general. But this is not something we have today. It would also be only one tool in reducing risk; it is a complement to guardrails on AI systems, not a substitute.

The best objection is one that I've rarely seen raised: that there is a gap between the models being useful in principle and the actual propensity of bad actors to use them. Most individual bad actors are disturbed individuals, so almost by definition their behavior is unpredictable and irrational—and it's *these* bad actors, the unskilled ones, who might have stood to benefit the most from AI making it much easier to kill many people. [24] Just because a type of violent attack is possible, doesn't mean someone will decide to do it. Perhaps biological attacks will be unappealing because they are reasonably likely to infect the perpetrator, they don't cater to the military-style fantasies that many violent individuals or groups have, and it is hard to selectively target

specific people. It could also be that going through a process that takes months, even if an AI walks you through it, involves an amount of patience that most disturbed individuals simply don't have. We may simply get lucky and motive and ability don't combine, in practice, in quite the right way.

But this seems like very flimsy protection to rely on. The motives of disturbed loners can change for any reason or no reason, and in fact there are already instances of LLMs being used in attacks (just not with biology). The focus on disturbed loners also ignores ideologically motivated terrorists, who are often willing to expend large amounts of time and effort (for example, the 9/11 hijackers). Wanting to kill as many people as possible is a motive that will probably arise sooner or later, and it unfortunately suggests bioweapons as the method. Even if this motive is extremely rare, it only has to materialize once. And as biology advances (increasingly driven by AI itself), it may also become possible to carry out more selective attacks (for example, targeted against people with specific ancestries), which adds yet another, very chilling, possible motive.

I do not think biological attacks will necessarily be carried out the instant it becomes widely possible to do so—in fact, I would bet against that. But added up across millions of people and a few years of time, I think there is a serious risk of a major attack, and the consequences would be so severe (with casualties potentially in the millions or more) that I believe we have no choice but to take serious measures to prevent it.

Defenses

That brings us to how to defend against these risks. Here I see three things we can do. First, AI companies can put guardrails on their models to prevent them from helping to produce bioweapons. Anthropic is very actively doing this. Claude's Constitution, which

mostly focuses on high-level principles and values, has a small number of specific hard-line prohibitions, and one of them relates to helping with the production of biological (or chemical, or nuclear, or radiological) weapons. But all models can be jailbroken, and so as a second line of defense, we've implemented (since mid-2025, when our tests showed our models were starting to get close to the threshold where they might begin to pose a risk) a classifier that specifically detects and blocks bioweapon-related outputs. We regularly upgrade and improve these classifiers, and have generally found them highly robust even against sophisticated adversarial attacks. [25] These classifiers increase the costs to serve our models measurably (in some models, they are close to 5% of total inference costs) and thus cut into our margins, but we feel that using them is the right thing to do.

To their credit, some other AI companies have implemented classifiers as well. But not every company has, and there is also nothing requiring companies to keep their classifiers. I am concerned that over time there may be a prisoner's dilemma where companies can defect and lower their costs by removing classifiers. This is once again a classic negative externalities problem that can't be solved by the voluntary actions of Anthropic or any other single company alone. [26] Voluntary industry standards may help, as may third-party evaluations and verification of the type done by AI security institutes and third-party evaluators.

But ultimately defense may require government action, which is the second thing we can do. My views here are the same as they are for addressing autonomy risks: we should start with transparency requirements, [27] which help society measure, monitor, and collectively defend against risks without disrupting economic activity in a heavy-handed way. Then, if and when we reach clearer thresholds of risk, we can craft legislation that more precisely targets these risks and has a lower chance of collateral damage. In the particular case of bioweapons, I actually think that the time for such targeted legislation may be

approaching soon—Anthropic and other companies are learning more and more about the nature of biological risks and what is reasonable to require of companies in defending against them. Fully defending against these risks may require working internationally, even with geopolitical adversaries, but there is precedent in treaties prohibiting the development of biological weapons. I am generally a skeptic about most kinds of international cooperation on AI, but this may be one narrow area where there is some chance of achieving global restraint. Even dictatorships do not want massive bioterrorist attacks.

Finally, the third countermeasure we can take is to try to develop defenses against biological attacks themselves. This could include monitoring and tracking for early detection, investments in air purification R&D (such as far-UVC disinfection), rapid vaccine development that can respond and adapt to an attack, better personal protective equipment (PPE), [28] and treatments or vaccinations for some of the most likely biological agents. mRNA vaccines, which can be designed to respond to a particular virus or variant, are an early example of what is possible here. Anthropic is excited to work with biotech and pharmaceutical companies on this problem. But unfortunately I think our expectations on the defensive side should be limited. There is an asymmetry between attack and defense in biology, because agents spread rapidly on their own, while defenses require detection, vaccination, and treatment to be organized across large numbers of people very quickly in response. Unless the response is lightning quick (which it rarely is), much of the damage will be done before a response is possible. It is conceivable that future technological improvements could shift this balance in favor of defense (and we should certainly use AI to help develop such technological advances), but until then, preventative safeguards will be our main line of defense.

It's worth a brief mention of cyberattacks here, since unlike biological attacks, AI-led cyberattacks have actually happened in the wild,

including at a large scale and for state-sponsored espionage. We expect these attacks to become more capable as models advance rapidly, until they are the main way in which cyberattacks are conducted. I expect AI-led cyberattacks to become a serious and unprecedented threat to the integrity of computer systems around the world, and Anthropic is working very hard to shut down these attacks and eventually reliably prevent them from happening. The reason I haven't focused on cyber as much as biology is that (1) cyberattacks are much less likely to kill people, certainly not at the scale of biological attacks, and (2) the offense-defense balance may be more tractable in cyber, where there is at least some hope that defense could keep up with (and even ideally outpace) AI attack if we invest in it properly.

Although biology is currently the most serious vector of attack, there are many other vectors and it is possible that a more dangerous one may emerge. The general principle is that without countermeasures, AI is likely to continuously lower the barrier to destructive activity on a larger and larger scale, and humanity needs a serious response to this threat.

3. The odious apparatus

Misuse for seizing power

The previous section discussed the risk of individuals and small organizations co-opting a small subset of the "country of geniuses in a datacenter" to cause large-scale destruction. But we should also worry—likely substantially more so—about misuse of AI for the purpose of *wielding or seizing power*, likely by larger and more established actors. [29]

In *Machines of Loving Grace*, I discussed the possibility that authoritarian governments might use powerful AI to surveil or repress their citizens in ways that would be extremely difficult to reform or

overthrow. Current autocracies are limited in how repressive they can be by the need to have humans carry out their orders, and humans often have limits in how inhumane they are willing to be. But AI-enabled autocracies would not have such limits.

Worse yet, countries could also use their advantage in AI to gain power over *other countries*. If the "country of geniuses" as a whole was simply owned and controlled by a single (human) country's military apparatus, and other countries did not have equivalent capabilities, it is hard to see how they could defend themselves: they would be outsmarted at every turn, similar to a war between humans and mice. Putting these two concerns together leads to the alarming possibility of a global totalitarian dictatorship. Obviously, it should be one of our highest priorities to prevent this outcome.

There are many ways in which AI could enable, entrench, or expand autocracy, but I'll list a few that I'm most worried about. Note that some of these applications have legitimate defensive uses, and I am not necessarily arguing against them in absolute terms; I am nevertheless worried that they structurally tend to favor autocracies:

- **Fully autonomous weapons.** A swarm of millions or billions of fully automated armed drones, locally controlled by powerful AI and strategically coordinated across the world by an even more powerful AI, could be an unbeatable army, capable of both defeating any military in the world and suppressing dissent within a country by following around every citizen. Developments in the Russia-Ukraine War should alert us to the fact that drone warfare is already with us (though not fully autonomous yet, and a tiny fraction of what might be possible with powerful AI). R&D from powerful AI could make the drones of one country far superior to those of others, speed up their manufacture, make them more resistant to electronic attacks, improve their maneuvering, and so

on. Of course, these weapons also have legitimate uses in the defense of democracy: they have been key to defending Ukraine and would likely be key to defending Taiwan. But they are a dangerous weapon to wield: we should worry about them in the hands of autocracies, but also worry that because they are so powerful, with so little accountability, there is a greatly increased risk of democratic governments turning them against their own people to seize power.

- **AI surveillance.** Sufficiently powerful AI could likely be used to compromise any computer system in the world, [30] and could also use the access obtained in this way to read *and make sense of* all the world's electronic communications (or even all the world's in-person communications, if recording devices can be built or commandeered). It might be frighteningly plausible to simply generate a complete list of anyone who disagrees with the government on any number of issues, even if such disagreement isn't explicit in anything they say or do. A powerful AI looking across billions of conversations from millions of people could gauge public sentiment, detect pockets of disloyalty forming, and stamp them out before they grow. This could lead to the imposition of a true panopticon on a scale that we don't see today, even with the CCP.
- **AI propaganda.** Today's phenomena of "AI psychosis" and "AI girlfriends" suggest that even at their current level of intelligence, AI models can have a powerful psychological influence on people. Much more powerful versions of these models, that were much more embedded in and aware of people's daily lives and could model and influence them over months or years, would likely be capable of essentially brainwashing many (most?) people into any desired ideology or attitude, and could be employed by an unscrupulous leader to ensure loyalty and suppress dissent, even in the face of a level of repression that most populations would

rebel against. Today people worry a lot about, for example, the potential influence of TikTok as CCP propaganda directed at children. I worry about that too, but a personalized AI agent that gets to know you over years and uses its knowledge of you to shape all of your opinions would be dramatically more powerful than this.

- **Strategic decision-making.** A country of geniuses in a datacenter could be used to advise a country, group, or individual on geopolitical strategy, what we might call a "virtual Bismarck." It could optimize the three strategies above for seizing power, plus probably develop many others that I haven't thought of (but that a country of geniuses could). Diplomacy, military strategy, R&D, economic strategy, and many other areas are all likely to be substantially increased in effectiveness by powerful AI. Many of these skills would be legitimately helpful for democracies—we want democracies to have access to the best strategies for defending themselves against autocracies—but the potential for misuse in *anyone's* hands still remains.

Having described *what* I am worried about, let's move on to *who*. I am worried about entities who have the most access to AI, who are starting from a position of the most political power, or who have an existing history of repression. In order of severity, I am worried about:

- **The CCP.** China is second only to the United States in AI capabilities, and is the country with the greatest likelihood of surpassing the United States in those capabilities. Their government is currently autocratic and operates a high-tech surveillance state. It has deployed AI-based surveillance already (including in the repression of Uyghurs), and is believed to employ algorithmic propaganda via TikTok (in addition to its many other international propaganda efforts). They have hands down the clearest path to the AI-enabled totalitarian nightmare I laid out

above. It may even be the default outcome within China, as well as within other autocratic states to whom the CCP exports surveillance technology. I have written often about the threat of the CCP taking the lead in AI and the existential imperative to prevent them from doing so. This is why. To be clear, I am not singling out China out of animus to them in particular—they are simply the country that most combines AI prowess, an autocratic government, and a high-tech surveillance state. If anything, it is the Chinese people themselves who are most likely to suffer from the CCP's AI-enabled repression, and they have no voice in the actions of their government. I greatly admire and respect the Chinese people and support the many brave dissidents within China and their struggle for freedom.

- **Democracies competitive in AI.** As I wrote above, democracies have a legitimate interest in some AI-powered military and geopolitical tools, because democratic governments offer the best chance to counter the use of these tools by autocracies. Broadly, I am supportive of arming democracies with the tools needed to defeat autocracies in the age of AI—I simply don't think there is any other way. But we cannot ignore the potential for abuse of these technologies by democratic governments themselves. Democracies normally have safeguards that prevent their military and intelligence apparatus from being turned inwards against their own population, [31] but because AI tools require so few people to operate, there is potential for them to circumvent these safeguards and the norms that support them. It is also worth noting that some of these safeguards are already gradually eroding in some democracies. Thus, we should arm democracies with AI, but we should do so carefully and within limits: they are the immune system we need to fight autocracies, but like the immune system, there is some risk of them turning on us and becoming a threat themselves.

- **Non-democratic countries with large datacenters.** Beyond China, most countries with less democratic governance are not leading AI players in the sense that they don't have companies which produce frontier AI models. Thus they pose a fundamentally different and lesser risk than the CCP, which remains the primary concern (most are also less repressive, and the ones that are more repressive, like North Korea, have no significant AI industry at all). But some of these countries do have large *datacenters* (often as part of buildouts by companies operating in democracies), which can be used to run frontier AI at large scale (though this does not confer the ability to push the frontier). There is some amount of danger associated with this—these governments could in principle expropriate the datacenters and use the country of AIs within it for their own ends. I am less worried about this compared to countries like China that directly develop AI, but it's a risk to keep in mind. [32]
- **AI companies.** It is somewhat awkward to say this as the CEO of an AI company, but I think the next tier of risk is actually AI companies themselves. AI companies control large datacenters, train frontier models, have the greatest expertise on how to use those models, and in some cases have daily contact with and the possibility of influence over tens or hundreds of millions of users. The main thing they lack is the legitimacy and infrastructure of a state, so much of what would be needed to build the tools of an AI autocracy would be illegal for an AI company to do, or at least exceedingly suspicious. But some of it is not impossible: they could, for example, use their AI products to brainwash their massive consumer user base, and the public should be alert to the risk this represents. I think the governance of AI companies deserves a lot of scrutiny.

There are a number of possible arguments against the severity of these threats, and I wish I believed them, because AI-enabled authoritarianism terrifies me. It's worth going through some of these arguments and responding to them.

First, some people might put their faith in the nuclear deterrent, particularly to counter the use of AI autonomous weapons for military conquest. If someone threatens to use these weapons against you, you can always threaten a nuclear response back. My worry is that I'm not totally sure we can be confident in the nuclear deterrent against a country of geniuses in a datacenter: it is possible that powerful AI could devise ways to detect and strike nuclear submarines, conduct influence operations against the operators of nuclear weapons infrastructure, or use AI's cyber capabilities to launch a cyberattack against satellites used to detect nuclear launches. [33] Alternatively, it's possible that taking over countries is feasible with only AI surveillance and AI propaganda, and never actually presents a clear moment where it's obvious what is going on and where a nuclear response would be appropriate. *Maybe* these things aren't feasible and the nuclear deterrent will still be effective, but it seems too high stakes to take a risk. [34]

A second possible objection is that there might be countermeasures we can take against these tools of autocracy. We can counter drones with our own drones, cyberdefense will improve along with cyberattack, there may be ways to immunize people against propaganda, etc. My response is that these defenses will only be possible with comparably powerful AI. If there isn't some counterforce with a comparably smart and numerous country of geniuses in a datacenter, it won't be possible to match the quality or quantity of drones, for cyberdefense to outsmart cyberoffense, etc. So the question of countermeasures reduces to the question of a balance of power in powerful AI. Here, I am concerned about the recursive or self-reinforcing property of powerful

AI (which I discussed at the beginning of this essay): that each generation of AI can be used to design and train the next generation of AI. This leads to a risk of a runaway advantage, where the current leader in powerful AI may be able to increase their lead and may be difficult to catch up with. We need to make sure it is not an authoritarian country that gets to this loop first.

Furthermore, even if a balance of power can be achieved, there is still risk that the world could be split up into autocratic spheres, as in *Nineteen Eighty-Four*. Even if several competing powers each have their powerful AI models, and none can overpower the others, each power could still internally repress their own population, and would be very difficult to overthrow (since the populations don't have powerful AI to defend themselves). It is thus important to prevent AI-enabled autocracy even if it doesn't lead to a single country taking over the world.

Defenses

How do we defend against this wide range of autocratic tools and potential threat actors? As in the previous sections, there are several things I think we can do. First, we should absolutely not be selling chips, chip-making tools, or datacenters to the CCP. Chips and chip-making tools are the single greatest bottleneck to powerful AI, and blocking them is a simple but extremely effective measure, perhaps the most important single action we can take. It makes no sense to sell the CCP the tools with which to build an AI totalitarian state and possibly conquer us militarily. A number of complicated arguments are made to justify such sales, such as the idea that "spreading our tech stack around the world" allows "America to win" in some general, unspecified economic battle. In my view, this is like selling nuclear weapons to North Korea and then bragging that the missile casings are made by Boeing and so the US is "winning." China is several years behind the

US in their ability to produce frontier chips in quantity, and the critical period for building the country of geniuses in a datacenter is very likely to be within those next several years. [35] There is no reason to give a giant boost to their AI industry during this critical period.

Second, it makes sense to use AI to empower democracies to resist autocracies. This is the reason Anthropic considers it important to provide AI to the intelligence and defense communities in the US and its democratic allies. Defending democracies that are under attack, such as Ukraine and (via cyber attacks) Taiwan, seems especially high priority, as does empowering democracies to use their intelligence services to disrupt and degrade autocracies from the inside. At some level the only way to respond to autocratic threats is to match and outclass them militarily. A coalition of the US and its democratic allies, if it achieved predominance in powerful AI, would be in a position to not only defend itself against autocracies, but contain them and limit their AI totalitarian abuses.

Third, we need to draw a hard line against AI abuses within democracies. There need to be limits to what we allow our governments to do with AI, so that they don't seize power or repress their own people. The formulation I have come up with is that we should use AI for national defense in all ways *except those which would make us more like our autocratic adversaries*.

Where should the line be drawn? In the list at the beginning of this section, two items—using AI for domestic mass surveillance and mass propaganda—seem to me like bright red lines and entirely illegitimate. Some might argue that there's no need to do anything (at least in the US), since domestic mass surveillance is already illegal under the Fourth Amendment. But the rapid progress of AI may create situations that our existing legal frameworks are not well designed to deal with. For example, it would likely not be unconstitutional for the US

government to conduct massively scaled recordings of all *public* conversations (e.g., things people say to each other on a street corner), and previously it would have been difficult to sort through this volume of information, but with AI it could all be transcribed, interpreted, and triangulated to create a picture of the attitude and loyalties of many or most citizens. I would support civil liberties-focused legislation (or maybe even a constitutional amendment) that imposes stronger guardrails against AI-powered abuses.

The other two items—fully autonomous weapons and AI for strategic decision-making—are harder lines to draw since they have legitimate uses in defending democracy, while also being prone to abuse. Here I think what is warranted is extreme care and scrutiny combined with guardrails to prevent abuses. My main fear is having too small a number of "fingers on the button," such that one or a handful of people could essentially operate a drone army without needing any other humans to cooperate to carry out their orders. As AI systems get more powerful, we may need to have more direct and immediate oversight mechanisms to ensure they are not misused, perhaps involving branches of government other than the executive. I think we should approach fully autonomous weapons in particular with great caution, [36] and not rush into their use without proper safeguards.

Fourth, after drawing a hard line against AI abuses in democracies, we should use that precedent to create an international taboo against the worst abuses of powerful AI. I recognize that the current political winds have turned against international cooperation and international norms, but this is a case where we sorely need them. The world needs to understand the dark potential of powerful AI in the hands of autocrats, and to recognize that certain uses of AI amount to an attempt to permanently steal their freedom and impose a totalitarian state from which they can't escape. I would even argue that in some cases, large-scale surveillance with powerful AI, mass propaganda with powerful AI,

and certain types of *offensive* uses of fully autonomous weapons should be considered crimes against humanity. More generally, a robust norm against AI-enabled totalitarianism and all its tools and instruments is sorely needed.

It is possible to have an even stronger version of this position, which is that because the possibilities of AI-enabled totalitarianism are so dark, autocracy is simply not a form of government that people can accept in the post-powerful AI age. Just as feudalism became unworkable with the industrial revolution, the AI age could lead inevitably and logically to the conclusion that democracy (and, hopefully, democracy improved and reinvigorated by AI, as I discuss in *Machines of Loving Grace*) is the only viable form of government if humanity is to have a good future.

Fifth and finally, AI companies should be carefully watched, as should their connection to the government, which is necessary, but must have limits and boundaries. The sheer amount of capability embodied in powerful AI is such that ordinary corporate governance—which is designed to protect shareholders and prevent ordinary abuses such as fraud—is unlikely to be up to the task of governing AI companies. There may also be value in companies publicly committing to (perhaps even as part of corporate governance) not take certain actions, such as privately building or stockpiling military hardware, using large amounts of computing resources by single individuals in unaccountable ways, or using their AI products as propaganda to manipulate public opinion in their favor.

The danger here comes from many directions, and some directions are in tension with others. The only constant is that we must seek accountability, norms, and guardrails for everyone, even as we empower "good" actors to keep "bad" actors in check.

4. Player piano

Economic disruption

The previous three sections were essentially about security risks posed by powerful AI: risks from the AI itself, risks from misuse by individuals and small organizations and risks of misuse by states and large organizations. If we put aside security risks or assume they have been solved, the next question is economic. What will be the effect of this infusion of incredible "human" capital on the economy? Clearly, the most obvious effect will be to greatly increase economic growth. The pace of advances in scientific research, biomedical innovation, manufacturing, supply chains, the efficiency of the financial system, and much more are almost guaranteed to lead to a much faster rate of economic growth. In *Machines of Loving Grace*, I suggest that a 10–20% sustained annual GDP growth rate may be possible.

But it should be clear that this is a double-edged sword: what are the economic prospects for most existing humans in such a world? New technologies often bring labor market shocks, and in the past humans have always recovered from them, but I am concerned that this is because these previous shocks affected only a small fraction of the full possible range of human abilities, leaving room for humans to expand to new tasks. AI will have effects that are much broader and occur much faster, and therefore I worry it will be much more challenging to make things work out well.

Labor market disruption

There are two specific problems I am worried about: labor market displacement, and concentration of economic power. Let's start with the first one. This is a topic that I warned about very publicly in 2025, where I predicted that AI could displace half of all entry-level white collar jobs in the next 1–5 years, even as it accelerates economic growth and scientific progress. This warning started a public debate about the topic. Many CEOs, technologists, and economists agreed with me, but

others assumed I was falling prey to a “lump of labor” fallacy and didn’t know how labor markets worked, and some didn’t see the 1–5-year time range and thought I was claiming AI is displacing jobs right now (which I agree it is likely not). So it is worth going through in detail why I am worried about labor displacement, to clear up these misunderstandings.

As a baseline, it’s useful to understand how labor markets *normally* respond to advances in technology. When a new technology comes along, it starts by making pieces of a given human job more efficient. For example, early in the Industrial Revolution, machines, such as upgraded plows, enabled human farmers to be more efficient at some aspects of the job. This improved the productivity of farmers, which increased their wages.

In the next step, some parts of the job of farming could be done *entirely* by machines, for example with the invention of the threshing machine or seed drill. In this phase, humans did a lower and lower fraction of the job, but the work they *did* complete became more and more leveraged because it is complementary to the work of machines, and their productivity continued to rise. As described by Jevons’ paradox, the wages of farmers and perhaps even the number of farmers continued to increase. Even when 90% of the job is being done by machines, humans can simply do 10x more of the 10% they still do, producing 10x as much output for the same amount of labor.

À terme, les machines prennent en charge la quasi-totalité des tâches, comme en témoignent les moissonneuses-batteuses , les tracteurs et autres équipements modernes. Dès lors, l'agriculture, en tant que forme d'emploi humain, connaît un déclin marqué, susceptible d'entraîner de graves perturbations à court terme. Cependant, l'agriculture n'étant qu'une des nombreuses activités utiles que les humains sont capables de réaliser, les individus finissent par se reconvertir, par exemple dans la

conduite de machines en usine. Ceci reste vrai même si l'agriculture représentait *autrefois* une part considérable de l'emploi . Il y a 250 ans, 90 % des Américains vivaient dans des fermes ; en Europe, 50 à 60 % de l'emploi était agricole . Aujourd'hui, ces pourcentages sont inférieurs à 10 % dans ces régions, car les travailleurs se sont tournés vers des emplois industriels (puis, plus tard, vers des emplois intellectuels). L'économie peut désormais accomplir ce qui nécessitait auparavant la majeure partie de la main-d'œuvre avec seulement 1 à 2 % de celle-ci, libérant ainsi le reste de la population active pour bâtir une société industrielle toujours plus avancée. Il n'existe plus de « volume de main-d'œuvre » fixe, mais une capacité sans cesse croissante à faire toujours plus avec toujours moins . Les salaires augmentent au même rythme que le PIB, et l'économie maintient le plein emploi une fois les perturbations à court terme passées.

Il est possible que les choses évoluent plus ou moins de la même manière avec l'IA, mais je parierais plutôt contre. Voici quelques raisons pour lesquelles je pense que l'IA sera probablement différente :

- **Vitesse.** Le rythme des progrès en IA est bien plus rapide que celui des précédentes révolutions technologiques. Par exemple, au cours des deux dernières années, les modèles d'IA sont passés de l'incapacité à exécuter une seule ligne de code à la capacité d' écrire la totalité ou la quasi-totalité du code pour certaines personnes, y compris des ingénieurs d'Anthropic. [37] Bientôt, ils pourront accomplir l'intégralité du travail d'un ingénieur logiciel, de bout en bout. [38] Il est difficile de s'adapter à ce rythme de changement, tant aux transformations des métiers qu'à la nécessité de se reconvertir. Même les programmeurs les plus réputés se disent de plus en plus dépassés. Ce rythme pourrait même s'accélérer, les modèles de programmation de l'IA accélérant sans cesse son développement. Précisons que cette accélération ne signifie pas que le marché du travail et l'emploi ne se redresseront pas à

terme ; elle implique simplement que la transition à court terme sera particulièrement difficile par rapport aux technologies précédentes, car les individus et le marché du travail sont lents à réagir et à retrouver leur équilibre.

- **Étendue cognitive.** Comme le suggère l'expression « pays de génies dans un centre de données », l'IA sera capable d'un très large éventail de capacités cognitives humaines, voire de toutes. Ceci est très différent des technologies précédentes comme l'agriculture mécanisée, les transports ou même les ordinateurs.[39] Cela compliquera la transition des individus des emplois disparus vers des postes similaires qui leur conviendraient. Par exemple, les compétences intellectuelles générales requises pour les postes de débutant dans la finance, le conseil et le droit sont assez similaires, même si les connaissances spécifiques diffèrent considérablement. Une technologie qui perturberait un seul de ces trois secteurs permettrait aux employés de se reconvertir vers les deux autres secteurs proches (ou aux étudiants de changer de spécialisation). Mais une perturbation simultanée des trois (ainsi que de nombreux autres emplois similaires) pourrait rendre l'adaptation plus difficile. De plus, il ne s'agit pas *seulement* de la disparition de la plupart des emplois existants. Ce phénomène s'est déjà produit : souvenez-vous que l'agriculture représentait une part importante de l'emploi. Or, les agriculteurs ont pu se reconvertir dans le travail relativement similaire d'opérateur de machines en usine, même si ce travail était peu courant auparavant. À l'inverse, l'IA se rapproche de plus en plus du profil cognitif général des humains, ce qui signifie qu'elle sera également performante dans les nouveaux emplois qui seraient créés suite à l'automatisation des anciens. Autrement dit, l'IA ne remplace pas des emplois humains spécifiques, mais plutôt une main-d'œuvre de manière générale.

- **Segmentation selon les capacités cognitives.** Pour un large éventail de tâches, l'IA semble progresser des niveaux les plus bas vers les plus élevés. Par exemple, en programmation, nos modèles sont passés du niveau de « codeur médiocre » à celui de « codeur compétent », puis à celui de « codeur très compétent ».^[40] Nous commençons à observer la même évolution dans le travail des cols blancs en général. Nous risquons donc de nous retrouver dans une situation où, au lieu d'affecter des personnes possédant des compétences spécifiques ou exerçant des professions particulières (qui peuvent s'adapter par une reconversion), l'IA affecte des personnes présentant certaines caractéristiques cognitives intrinsèques, à savoir des capacités intellectuelles plus faibles (plus difficiles à modifier). On ignore ce que l'avenir leur réserve et ce qu'ils feront, et je crains qu'elles ne forment une « sous-classe » de chômeurs ou de personnes très mal rémunérées. Certes, des situations similaires se sont déjà produites : par exemple, certains économistes considèrent que les ordinateurs et Internet représentent un « progrès technologique biaisé en faveur des compétences ». Mais ce biais n'était pas aussi marqué que ce que je prévois avec l'IA, et il aurait contribué à creuser les inégalités salariales.^[41] Ce n'est donc pas exactement un précédent rassurant.
- **Capacité à combler les lacunes.** Face aux nouvelles technologies, l'adaptation des emplois humains repose souvent sur la multiplicité de leurs aspects. Or, même si la nouvelle technologie semble remplacer directement l'humain, elle présente souvent des lacunes. Si l'on invente une machine à fabriquer des objets, l'alimentation en matières premières reste nécessaire. Même si cette tâche ne représente que 1 % de l'effort requis pour une fabrication manuelle, les ouvriers peuvent produire 100 fois plus d'objets. L'IA, en plus d'être une technologie en plein essor, est aussi une technologie *d'adaptation* rapide . À chaque nouvelle

version, les entreprises spécialisées évaluent avec précision les points forts et les faiblesses du modèle, et les clients fournissent également ces informations après le lancement. Les faiblesses peuvent être corrigées en collectant des tâches illustrant les lacunes actuelles et en entraînant le modèle suivant sur ces tâches. Au début de l'IA générative, les utilisateurs ont constaté certaines faiblesses des systèmes (comme des modèles d'images générant des mains avec un nombre de doigts incorrect) et beaucoup ont supposé que ces faiblesses étaient inhérentes à la technologie. Si tel était le cas, les perturbations sur le marché du travail seraient limitées. Mais la plupart de ces faiblesses sont rapidement corrigées, souvent en quelques mois seulement.

Il est important d'aborder les principaux points de scepticisme. Premièrement, on entend souvent dire que la diffusion économique sera lente, de sorte que même si la technologie sous-jacente est *capable* d'effectuer la majeure partie du travail humain, son application concrète à l'ensemble de l'économie pourrait être beaucoup plus lente (par exemple, dans des secteurs éloignés de l'IA et lents à adopter cette technologie). La lenteur de la diffusion technologique est indéniable : je discute avec des personnes issues d'entreprises très diverses, et il existe des endroits où l'adoption de l'IA prendra des années. C'est pourquoi je prévois que 50 % des emplois de cols blancs débutants seront bouleversés d'ici 1 à 5 ans, même si je pense que nous disposerons d'une IA puissante (qui serait, technologiquement parlant, suffisante pour effectuer *la plupart, voire la totalité,* des emplois, et pas seulement les emplois débutants) bien avant 5 ans. Mais les effets de diffusion ne font que nous donner du temps. Et je ne suis pas certain que ce sera aussi lent que certains le prévoient. L'adoption de l'IA en entreprise progresse à un rythme bien plus rapide que pour toute autre technologie auparavant, principalement grâce à la puissance intrinsèque de la technologie. De plus, même si les entreprises traditionnelles tardent à adopter les nouvelles technologies, des

startups surgiront pour faciliter cette adoption. Si cela ne fonctionne pas, elles pourraient tout simplement bouleverser directement le marché établi.

Cela pourrait mener à un monde où ce ne seraient pas tant des emplois spécifiques qui seraient bouleversés, mais plutôt les grandes entreprises dans leur ensemble, remplacées par des start-ups beaucoup moins gourmandes en main-d'œuvre. Il pourrait également en résulter une « inégalité géographique », où une part croissante de la richesse mondiale se concentrerait dans la Silicon Valley, qui deviendrait une économie à part, fonctionnant à un rythme différent du reste du monde et le laissant à la traîne. Tous ces scénarios seraient excellents pour la croissance économique, mais beaucoup moins pour le marché du travail et les populations laissées pour compte.

Deuxièmement, certains affirment que les emplois humains se déplaceront vers le monde physique, ce qui exclut toute la catégorie du « travail cognitif », domaine où l'IA progresse à une vitesse fulgurante. Je ne suis pas certain de la pertinence de cette affirmation. De nombreuses tâches physiques sont déjà effectuées par des machines (par exemple, dans le secteur manufacturier) ou le seront bientôt (par exemple, la conduite automobile). De plus, une IA suffisamment puissante sera capable d'accélérer le développement des robots, puis de les contrôler dans le monde physique. Cela pourrait nous faire gagner du temps (ce qui est une bonne chose), mais je crains que ce ne soit que très limité. Et même si cette perturbation se limitait aux tâches cognitives, elle n'en serait pas moins d'une ampleur et d'une rapidité sans précédent.

Troisièmement, certaines tâches requièrent peut-être intrinsèquement une intervention humaine, ou en tirent un grand bénéfice. J'ai quelques doutes à ce sujet, mais je reste sceptique quant à sa capacité à compenser la majeure partie des impacts décrits précédemment. L'IA

est déjà largement utilisée dans le service client. Nombreux sont ceux qui affirment qu'il est plus facile de parler de leurs problèmes personnels à une IA qu'à un thérapeute ; l'IA serait en effet plus patiente. Lorsque ma sœur a rencontré des difficultés médicales pendant sa grossesse, elle a eu le sentiment de ne pas obtenir les réponses ni le soutien nécessaires de la part des soignants, et elle a trouvé que Claude avait un meilleur contact humain (et qu'il réussissait mieux à diagnostiquer le problème). Je suis certain qu'il existe des tâches pour lesquelles le contact humain est essentiel, mais j'ignore combien elles sont nombreuses – et il s'agit ici de trouver un emploi pour la quasi-totalité des personnes sur le marché du travail.

Quatrièmement, certains pourraient arguer que l'avantage comparatif protège toujours les humains. Selon la loi de l'avantage comparatif , même si l'IA surpasse les humains en tout, toute différence *relative* entre les profils de compétences humains et IA crée une base d'échanges et de spécialisation entre eux. Le problème est que si les IA sont littéralement des milliers de fois plus productives que les humains, cette logique commence à s'effondrer. Même des coûts de transaction minimes pourraient rendre les échanges commerciaux entre IA et humains non rentables. Et les salaires humains pourraient être très bas, même s'ils ont techniquement une valeur ajoutée.

Il est possible que tous ces facteurs puissent être pris en compte et que le marché du travail soit suffisamment résilient pour s'adapter même à une perturbation d'une telle ampleur. Mais même s'il parvient à s'adapter, les facteurs évoqués laissent présager que le choc à court terme sera d'une ampleur sans précédent.

Défenses

Que pouvons-nous faire face à ce problème ? J’ai plusieurs suggestions, dont certaines sont déjà mises en œuvre par Anthropic. La première consiste simplement à obtenir des données précises et en temps réel sur

l'évolution des suppressions d'emplois. Lorsqu'une mutation économique survient rapidement, il est difficile d'obtenir des données fiables, et sans données fiables, il est difficile d'élaborer des politiques efficaces. Par exemple, les données gouvernementales manquent actuellement de données fines et fréquentes sur l'adoption de l'IA dans les entreprises et les secteurs. Depuis un an, Anthropic exploite et publie un Indice économique qui présente l'utilisation de nos modèles quasiment en temps réel, ventilée par secteur, tâche, lieu, et même selon que la tâche soit automatisée ou réalisée en collaboration. Nous disposons également d'un Conseil consultatif économique pour nous aider à interpréter ces données et à anticiper les tendances.

Deuxièmement, les entreprises spécialisées en IA ont le choix quant à leur collaboration avec les entreprises. L'inefficacité même des entreprises traditionnelles implique que leur déploiement de l'IA peut être fortement tributaire de leur stratégie, et il est possible d'opter pour une approche plus efficace. Les entreprises doivent souvent choisir entre « réduire les coûts » (faire la même chose avec moins de personnel) et « innover » (faire plus avec le même effectif). Le marché finira inévitablement par proposer les deux, et toute entreprise d'IA compétitive devra répondre à ces deux besoins. Toutefois, il est possible d'orienter les entreprises vers l'innovation lorsque cela est envisageable, ce qui pourrait nous faire gagner du temps. Anthropic se penche activement sur cette question.

Troisièmement, les entreprises devraient réfléchir à la manière de prendre soin de leurs employés. À court terme, faire preuve de créativité quant aux redéploiements internes pourrait s'avérer une solution prometteuse pour éviter les licenciements. À long terme, dans un monde où la richesse totale est immense et où de nombreuses entreprises voient leur valeur augmenter considérablement grâce à l'accroissement de la productivité et à la concentration des capitaux, il pourrait devenir envisageable de rémunérer les employés même

longtemps après qu'ils aient cessé d'apporter une valeur économique au sens traditionnel du terme. Anthropic étudie actuellement différentes pistes pour ses propres employés, que nous partagerons prochainement.

Quatrièmement, les personnes fortunées ont l'obligation de contribuer à la résolution de ce problème. Je regrette que nombre d'entre elles (notamment dans le secteur technologique) aient récemment adopté une attitude cynique et nihiliste, considérant la philanthropie comme inévitablement frauduleuse ou inutile. Or, la philanthropie privée, à l'instar de la Fondation Gates , et les programmes publics, comme le PEPFAR, ont sauvé des dizaines de millions de vies dans les pays en développement et contribué à créer des opportunités économiques dans les pays développés. Tous les cofondateurs d'Anthropic se sont engagés à donner 80 % de leur fortune, et les employés d'Anthropic se sont engagés individuellement à donner des actions de l'entreprise d'une valeur de plusieurs milliards au cours actuel – des dons que l'entreprise s'est engagée à égaler.

Cinquièmement, bien que toutes les actions privées mentionnées ci-dessus puissent s'avérer utiles, un problème macroéconomique d'une telle ampleur nécessitera en fin de compte une intervention gouvernementale. La réponse politique naturelle à une économie florissante, conjuguée à de fortes inégalités (dues au manque d'emplois ou à des emplois mal rémunérés pour beaucoup), est une fiscalité progressive. Cette taxe pourrait être générale ou cibler spécifiquement les entreprises spécialisées en intelligence artificielle. Bien entendu, la conception fiscale est complexe et les risques d'erreur sont nombreux. Je ne soutiens pas les politiques fiscales mal conçues. J'estime que les niveaux extrêmes d'inégalité prédits dans cet essai justifient, pour des raisons morales fondamentales, une politique fiscale plus rigoureuse. Cependant, je peux également avancer un argument pragmatique aux milliardaires du monde entier : il est dans leur intérêt de soutenir une

version efficace de cette politique. S'ils ne la soutiennent pas, ils se retrouveront inévitablement avec une version inefficace, conçue par une majorité.

En fin de compte, je considère toutes les interventions mentionnées ci-dessus comme des moyens de gagner du temps. À terme, l'IA sera capable de tout faire, et nous devons nous y préparer. J'espère que d'ici là, nous pourrons utiliser l'IA elle-même pour restructurer les marchés de manière à ce que cela profite à tous, et que les interventions évoquées ci-dessus nous permettront de traverser cette période de transition.

Concentration économique du pouvoir

Indépendamment du problème des pertes d'emplois ou des inégalités économiques *en tant que telles,* se pose celui de la *concentration du pouvoir économique.* La section 1 a abordé le risque de dépossession de l'humanité par l'IA, et la section 3, celui de la dépossession des citoyens par leurs gouvernements, par la force ou la coercition. Mais une autre forme de dépossession peut survenir en cas de concentration si importante des richesses qu'un petit groupe d'individus contrôle de fait la politique gouvernementale grâce à son influence, tandis que les citoyens ordinaires sont impuissants, faute de levier économique. La démocratie repose fondamentalement sur l'idée que la population dans son ensemble est indispensable au fonctionnement de l'économie. Si ce levier économique disparaît, le contrat social implicite de la démocratie risque de ne plus fonctionner. D'autres ont déjà traité de ce sujet , je n'ai donc pas besoin de m'y attarder, mais je partage cette inquiétude et je crains que ce phénomène ne soit déjà en train de se produire.

Soyons clairs : je ne suis pas opposé à ce que certaines personnes gagnent beaucoup d’argent. On peut même affirmer que, dans des conditions normales, cela stimule la croissance économique. Je comprends les inquiétudes liées au risque de freiner l’innovation en

tuant la poule aux œufs d'or qui la génère. Mais dans un scénario où la croissance du PIB atteint 10 à 20 % par an et où l'IA s'impose rapidement dans l'économie, alors même que des individus détiennent une part importante du PIB, l'innovation n'est *pas* le problème. Le problème, c'est la concentration des richesses, qui risque de déstabiliser la société.

L'exemple le plus célèbre de concentration extrême des richesses dans l'histoire des États-Unis est l' Âge d'or américain , et l'industriel le plus riche de cette période était John D. Rockefeller . La fortune de Rockefeller représentait environ 2 % du PIB américain de l'époque. Une part similaire de ces richesses aujourd'hui représenterait une fortune de 600 milliards de dollars, et l'homme le plus riche du monde (Elon Musk) dépasse déjà ce montant, avec environ 700 milliards de dollars . Nous atteignons donc déjà des niveaux de concentration des richesses sans précédent dans l'histoire, *avant* même que l'IA n'ait pleinement impacté notre économie. Il n'est pas exagéré, à mon avis, d'imaginer (dans le cas d'un « pays de génies ») que les entreprises d'IA, les fabricants de semi-conducteurs et peut-être les entreprises d'applications en aval génèrent un chiffre d'affaires d'environ 3 000 milliards de dollars par an . Avec une valeur estimée à environ 30 000 milliards de dollars, et des fortunes personnelles se chiffrant en milliers de milliards, les débats actuels sur la politique fiscale n'auront plus lieu d'être, la situation étant fondamentalement différente.

À cet égard, le couplage de cette concentration économique des richesses avec le système politique m'inquiète déjà. Les centres de données d'IA représentent déjà une part importante de la croissance économique américaine, [44] Ces mesures lient étroitement les intérêts financiers des grandes entreprises technologiques (de plus en plus axées sur l'IA ou son infrastructure) et les intérêts politiques du gouvernement, ce qui peut engendrer des incitations perverses. On le constate déjà à travers la réticence des entreprises technologiques à

critiquer le gouvernement américain et le soutien de ce dernier à des politiques de déréglementation de l'IA extrêmement restrictives.

Défenses

Que peut-on faire ? Premièrement, et c'est évident, les entreprises devraient tout simplement choisir de ne pas y participer. Anthropic s'est toujours efforcé d'être un acteur de l'élaboration des politiques et non un acteur politique, et de maintenir ses positions authentiques quelle que soit l'administration en place. Nous nous sommes prononcés en faveur d' une réglementation raisonnable de l'IA et de contrôles à l'exportation conformes à l'intérêt public, même lorsque ceux-ci sont en contradiction avec la politique gouvernementale. Beaucoup de gens m'ont dit que nous devrions arrêter de faire cela, que cela pourrait entraîner un traitement défavorable, mais au cours de l'année où nous l'avons fait, la valorisation d'Anthropic a été multipliée par plus de 6, une augmentation presque sans précédent à notre échelle commerciale.

Deuxièmement, le secteur de l'IA a besoin d'une relation plus saine avec les pouvoirs publics, fondée sur un dialogue constructif autour des politiques publiques plutôt que sur un simple alignement politique. Notre choix de privilégier le fond des débats politiques est parfois perçu comme une erreur tactique ou un manque de discernement, plutôt que comme une décision de principe, et cette perception m'inquiète. Dans une démocratie saine, les entreprises devraient pouvoir défendre des politiques pertinentes pour elles-mêmes. Par ailleurs, une réaction négative du public envers l'IA se dessine : elle pourrait être salutaire, mais elle **manque** actuellement de clarté. Une grande partie de cette réaction s'attaque à des problèmes qui n'en sont pas vraiment (comme la consommation d'eau des centres de données) et propose des solutions (comme l'interdiction des centres de données ou des taxes sur la fortune mal conçues) qui ne répondent pas aux véritables préoccupations. Le problème de fond qui mérite notre

attention est de garantir que le développement de l'IA reste au service de l'intérêt public, sans être instrumentalisé par une quelconque alliance politique ou commerciale. Il me semble donc essentiel de concentrer le débat public sur ce point.

Troisièmement, les interventions macroéconomiques que j'ai décrites précédemment dans cette section, ainsi qu'un regain de philanthropie privée, peuvent contribuer à rééquilibrer l'économie, en s'attaquant simultanément aux problèmes de suppression d'emplois et de concentration du pouvoir économique. Il convient de se référer à l'histoire de notre pays : même à l'âge d'or américain, des industriels comme Rockefeller et Carnegie ressentaient une forte obligation envers la société, le sentiment que celle-ci avait largement contribué à leur succès et qu'ils se devaient de lui rendre la pareille. Cet esprit semble aujourd'hui se faire de plus en plus rare, et je pense qu'il constitue une part importante de la solution à ce dilemme économique. Ceux qui sont à l'avant-garde de l'essor économique de l'IA devraient être prêts à partager leur richesse et leur pouvoir.

5. Mers noires de l'infini

Effets indirects

Cette dernière section englobe toutes les inconnues inconnues, notamment les problèmes qui pourraient survenir indirectement suite aux progrès de l'IA et à l'accélération générale des sciences et des technologies qui en découle. Supposons que nous maîtrisions tous les risques décrits jusqu'ici et commencions à bénéficier des avantages de l'IA. Nous assisterons probablement à un « siècle de progrès scientifique et économique condensé en une décennie », ce qui sera extrêmement positif pour le monde. Cependant, nous devrons ensuite faire face aux problèmes engendrés par ce rythme de progrès effréné, et ces problèmes pourraient survenir rapidement. Nous pourrions

également rencontrer d'autres risques, conséquences indirectes des progrès de l'IA, difficiles à anticiper.

Compte tenu de la nature des inconnues inconnues, il est impossible d'établir une liste exhaustive, mais je citerai trois points à surveiller à titre d'exemples illustratifs :

- **Les progrès rapides de la biologie.** Si nous parvenons à réaliser un siècle de progrès médicaux en quelques années, il est possible que nous augmentions considérablement l'espérance de vie humaine et que nous acquérions des capacités radicales, comme celle d'accroître l'intelligence humaine ou de modifier profondément la biologie humaine. Ce seraient des changements majeurs dans le champ des possibles, qui se produiraient très rapidement. Ils pourraient être positifs s'ils sont menés de manière responsable (ce que j'espère, comme je l'explique dans *Machines of Loving Grace*), mais le risque qu'ils tournent mal est toujours présent – par exemple, si les efforts pour rendre les humains plus intelligents les rendent aussi plus instables ou avides de pouvoir. Il y a aussi la question des « téléchargements » ou de l'« émulation cérébrale complète », des esprits humains numériques incarnés dans un logiciel, qui pourraient un jour aider l'humanité à transcender ses limites physiques, mais qui comportent également des risques que je trouve inquiétants .
- **L'IA transforme la vie humaine de manière néfaste.** Un monde peuplé de milliards d'intelligences bien plus performantes que l'humain dans tous les domaines sera un monde pour le moins étrange. Même si l'IA ne vise pas activement à attaquer les humains (Section 1) et n'est pas explicitement utilisée par les États à des fins d'oppression ou de contrôle (Section 3), de nombreux problèmes pourraient survenir, notamment via des incitations commerciales classiques et des transactions en apparence consensuelles. On en perçoit déjà les premiers signes dans les

inquiétudes concernant la psychose induite par l'IA, les suicides provoqués par l'IA et les relations amoureuses avec des IA. Par exemple, des IA puissantes pourraient-elles inventer une nouvelle religion et convertir des millions de personnes ? La plupart des gens pourraient-ils devenir dépendants des interactions avec l'IA ? Pourrions-nous finir par être manipulés par des systèmes d'IA, qui surveilleraient chacun de nos mouvements et nous dicteraient nos actions et nos paroles en permanence, nous menant à une vie « satisfaisante » certes, mais dépourvue de liberté et de toute fierté du travail accompli ? Il serait facile de générer des dizaines de scénarios de ce genre si je m'asseyais avec le créateur de *Black Mirror* et que nous essayions de les imaginer ensemble. Je pense que cela souligne l'importance d'améliorer la Constitution de Claude , au-delà de ce qui est nécessaire pour prévenir les problèmes évoqués dans la section 1. Il me semble crucial de veiller à ce que les modèles d'IA prennent *réellement* en compte les intérêts à long terme de leurs utilisateurs, d'une manière que des personnes sensées approuveraient plutôt que d'une façon subtilement déformée.

- **Human purpose.** This is related to the previous point, but it's not so much about specific human interactions with AI systems as it is about how human life changes in general in a world with powerful AI. Will humans be able to find purpose and meaning in such a world? I think this is a matter of attitude: as I said in *Machines of Loving Grace*, I think human purpose does not depend on being the best in the world at something, and humans can find purpose even over very long periods of time through stories and projects that they love. We simply need to break the link between the generation of economic value and self-worth and meaning. But that is a transition society has to make, and there is always the risk we don't handle it well.

My hope with all of these potential problems is that in a world with powerful AI that we trust not to kill us, that is not the tool of an oppressive government, and that is genuinely working on our behalf, we can use AI itself to anticipate and prevent these problems. But that is not guaranteed—like all of the other risks, it is something we have to handle with care.

Humanity's test

Reading this essay may give the impression that we are in a daunting situation. I certainly found it daunting to write, in contrast with *Machines of Loving Grace,* which felt like giving form and structure to surpassingly beautiful music that had been echoing in my head for years. And there is much about the situation that genuinely is hard. AI brings threats to humanity from multiple directions, and there is genuine tension between the different dangers, where mitigating some of them risks making others worse if we do not thread the needle extremely carefully.

Taking time to carefully build AI systems so they do not autonomously threaten humanity is in genuine tension with the need for democratic nations to stay ahead of authoritarian nations and not be subjugated by them. But in turn, the same AI-enabled tools that are necessary to fight autocracies can, if taken too far, be turned inward to create tyranny in our own countries. AI-driven terrorism could kill millions through the misuse of biology, but an overreaction to this risk could lead us down the road to an autocratic surveillance state. The labor and economic concentration effects of AI, in addition to being grave problems in their own right, may force us to face the other problems in an environment of public anger and perhaps even civil unrest, rather than being able to call on the better angels of our nature. Above all, the sheer *number* of

risks, including unknown ones, and the need to deal with all of them at once, creates an intimidating gauntlet that humanity must run.

Furthermore, the last few years should make clear that the idea of stopping or even substantially slowing the technology is fundamentally untenable. The formula for building powerful AI systems is incredibly simple, so much so that it can almost be said to emerge spontaneously from the right combination of data and raw computation. Its creation was probably inevitable the instant humanity invented the transistor, or arguably even earlier when we first learned to control fire. If one company does not build it, others will do so nearly as fast. If all companies in democratic countries stopped or slowed development, by mutual agreement or regulatory decree, then authoritarian countries would simply keep going. Given the incredible economic and military value of the technology, together with the lack of any meaningful enforcement mechanism, I don't see how we could possibly convince them to stop.

I do see a path to a *slight* moderation in AI development that is compatible with a realist view of geopolitics. That path involves slowing down the march of autocracies towards powerful AI for a few years by denying them the resources they need to build it, [46] namely chips and semiconductor manufacturing equipment. This in turn gives democratic countries a buffer that they can "spend" on building powerful AI more carefully, with more attention to its risks, while still proceeding fast enough to comfortably beat the autocracies. The race between AI companies within democracies can then be handled under the umbrella of a common legal framework, via a mixture of industry standards and regulation.

Anthropic has advocated very hard for this path, by pushing for chip export controls and judicious regulation of AI, but even these seemingly common-sense proposals have largely been rejected by

policymakers in the United States (which is the country where it's most important to have them). There is so much money to be made with AI—literally trillions of dollars per year—that even the simplest measures are finding it difficult to overcome the political economy inherent in AI. This is the trap: AI is so powerful, such a glittering prize, that it is very difficult for human civilization to impose any restraints on it at all.

I can imagine, as Sagan did in *Contact*, that this same story plays out on thousands of worlds. A species gains sentience, learns to use tools, begins the exponential ascent of technology, faces the crises of industrialization and nuclear weapons, and if it survives those, confronts the hardest and final challenge when it learns how to shape sand into machines that think. Whether we survive that test and go on to build the beautiful society described in *Machines of Loving Grace*, or succumb to slavery and destruction, will depend on our character and our determination as a species, our spirit and our soul.

Despite the many obstacles, I believe humanity has the strength inside itself to pass this test. I am encouraged and inspired by the thousands of researchers who have devoted their careers to helping us understand and steer AI models, and to shaping the character and constitution of these models. I think there is now a good chance that those efforts bear fruit in time to matter. I am encouraged that at least some companies have stated they'll pay meaningful commercial costs to block their models from contributing to the threat of bioterrorism. I am encouraged that a few brave people have resisted the prevailing political winds and passed legislation that puts the first early seeds of sensible guardrails on AI systems. I am encouraged that the public understands that AI carries risks and wants those risks addressed. I am encouraged by the indomitable spirit of freedom around the world and the determination to resist tyranny wherever it occurs.

But we will need to step up our efforts if we want to succeed. The first step is for those closest to the technology to simply tell the truth about the situation humanity is in, which I have always tried to do; I'm doing so more explicitly and with greater urgency with this essay. The next step will be convincing the world's thinkers, policymakers, companies, and citizens of the imminence and overriding importance of this issue—that it is worth expending thought and political capital on this in comparison to the thousands of other issues that dominate the news every day. Then there will be a time for courage, for enough people to buck the prevailing trends and stand on principle, even in the face of threats to their economic interests and personal safety.

The years in front of us will be impossibly hard, asking more of us than we think we can give. But in my time as a researcher, leader, and citizen, I have seen enough courage and nobility to believe that we can win—that when put in the darkest circumstances, humanity has a way of gathering, seemingly at the last minute, the strength and wisdom needed to prevail. We have no time to lose.

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Footnotes

[1] This is symmetric to a point I made in *Machines of Loving Grace*, where I started by saying that AI's upsides shouldn't be thought of in terms of a prophecy of salvation, and that it's important to be concrete

and grounded and to avoid grandiosity. Ultimately, prophecies of salvation and prophecies of doom are unhelpful for confronting the real world, for basically the same reasons. ↩

[2] Anthropic's goal is to remain consistent through such changes. When talking about AI risks was politically popular, Anthropic cautiously advocated for a judicious and evidence-based approach to these risks. Now that talking about AI risks is politically unpopular, Anthropic continues to cautiously advocate for a judicious and evidence-based approach to these risks. ↩

[3] Over time, I have gained increasing confidence in the trajectory of AI and the likelihood that it will surpass human ability across the board, but some uncertainty still remains. ↩

[4] Export controls for chips are a great example of this. They are simple and appear to mostly just work. ↩

[5] And of course, the hunt for such evidence must be intellectually honest, such that it could also turn up evidence of a lack of danger. Transparency through model cards and other disclosures is an attempt at such an intellectually honest endeavor. ↩

[6] Indeed, since writing *Machines of Loving Grace* in 2024, AI systems have become capable of doing tasks that take humans several hours, with METR recently assessing that Opus 4.5 can do about four human hours of work with 50% reliability. ↩

[7] And to be clear, even if powerful AI is only 1–2 years away in a technical sense, many of its societal consequences, both positive and negative, may take a few years longer to occur. This is why I can simultaneously think that AI will disrupt 50% of *entry-level* white-collar jobs over 1–5 years, while also thinking we may have AI that is more capable than *everyone* in only 1–2 years. ↩

[8] It is worth adding that the *public* (as compared to policymakers) does seem to be very concerned with AI risks. I think some of their focus is correct (i.e. AI job displacement), and some is misguided (such as concerns about water use of AI, which is not significant). This backlash gives me hope that a consensus around addressing risks is possible, but so far it has not yet been translated into policy changes, let alone effective or well-targeted policy changes. ↩

[9] They can also, of course, manipulate (or simply pay) large numbers of humans into doing what they want in the physical world. ↩

[10] I don't think this is a straw man: it's my understanding, for example, that Yann LeCun holds this position. ↩

[11] For example, see Section 5.5.2 (p. 63–66) of the Claude 4 system card. ↩

[12] There are also a number of other assumptions inherent in the simple model, which I won't discuss here. Broadly, they should make us less worried about the specific simple story of misaligned power-seeking, but also more worried about possible unpredictable behavior we haven't anticipated. ↩

[13] *Ender's Game* describes a version of this involving humans rather than AI. ↩

[14] For example, models may be told not to do various bad things, and also to obey humans, but may then observe that many humans do exactly those bad things! It's not clear how this contradiction would resolve (and a well-designed constitution should encourage the model to handle these contradictions gracefully), but this type of dilemma is not so different from the supposedly "artificial" situations that we put AI models in during testing. ↩

[15] Incidentally, one consequence of the constitution being a natural-language document is that it is legible to the world, and that means it can be critiqued by anyone and compared to similar documents by other companies. It would be valuable to create a race to the top that not only encourages companies to release these documents, but encourages them to be good. ↩

[16] There's even a hypothesis about a deep unifying principle connecting the character-based approach from Constitutional AI to results from interpretability and alignment science. According to the hypothesis, the fundamental mechanisms driving Claude originally arose as ways for it to simulate characters in pretraining, such as predicting what the characters in a novel would say. This would suggest that a useful way to think about the constitution is more like a character description that the model uses to instantiate a consistent persona. It would also help us explain the "I must be a bad person" results I mentioned above (because the model is trying to *act as if* it's a coherent character—in this case a bad one), and would suggest that interpretability methods should be able to discover "psychological traits" within models. Our researchers are working on ways to test this hypothesis. ↩

[17] To be clear, monitoring is done in a privacy-preserving way. ↩

[18] Even in our own experiments with what are essentially voluntarily imposed rules with our Responsible Scaling Policy, we have found over and over again that it's very easy to end up being too rigid, by drawing lines that seem important ex ante but turn out to be silly in retrospect. It is just very easy to set rules about the wrong things when a technology is advancing rapidly. ↩

[19] SB 53 and RAISE do not apply at all to companies with under $500M in annual revenue. They only apply to larger, more established companies like Anthropic. ↩

[20] I originally read Joy's essay 25 years ago, when it was written, and it had a profound impact on me. Then and now, I do see it as too pessimistic—I don't think broad "relinquishment" of whole areas of technology, which Joy suggests, is the answer—but the issues it raises were surprisingly prescient, and Joy also writes with a deep sense of compassion and humanity that I admire. ↩

[21] We do have to worry about state actors, now and in the future, and I discuss that in the next section. ↩

[22] There is evidence that many terrorists are at least relatively well-educated, which might seem to contradict what I'm arguing here about a negative correlation between ability and motivation. But I think in actual fact they are compatible observations: if the ability threshold for a successful attack is high, then almost by definition those who *currently* succeed must have high ability, even if ability and motivation are negatively correlated. But in a world where the limitations on ability were removed (e.g., with future LLMs), I'd predict that a substantial population of people with the motivation to kill but lower ability would start to do so—just as we see for crimes that don't require much ability (like school shootings). ↩

[23] Aum Shinrikyo did try, however. The leader of Aum Shinrikyo, Seiichi Endo, had training in virology from Kyoto University, and attempted to produce both anthrax and ebola. However, as of 1995, even he lacked enough expertise and resources to succeed at this. The bar is now substantially lower, and LLMs could reduce it even further. ↩

[24] A bizarre phenomenon relating to mass murderers is that the style of murder they choose operates almost as a grotesque sort of fad. In the 1970s and 1980s, serial killers were very common, and new serial killers often copied the behavior of more established or famous serial killers. In the 1990s and 2000s, mass shootings became more common,

while serial killers became less common. There is no technological change that triggered these patterns of behavior, it just appears that violent murderers were copying each others' behavior and the "popular" thing to copy changed. ↩

[25] Casual jailbreakers sometimes believe that they've compromised these classifiers when they get the model to output one specific piece of information, such as the genome sequence of a virus. But as I explained before, the threat model we are worried about involves step-by-step, interactive advice that extends over weeks or months about specific obscure steps in the bioweapons production process, and this is what our classifiers aim to defend against. (We often describe our research as looking for "universal" jailbreaks—ones that don't just work in one specific or narrow context, but broadly open up the model's behavior.) ↩

[26] Though we will continue to invest in work to make our classifiers more efficient, and it may make sense for companies to share advances like these with one another. ↩

[27] Obviously, I do not think companies should have to disclose technical details about the specific steps in biological weapons production that they are blocking, and the transparency legislation that has been passed so far (SB 53 and RAISE) accounts for this issue. ↩

[28] Another related idea is "resilience markets" where the government encourages stockpiling of PPE, respirators, and other essential equipment needed to respond to a biological attack by promising ahead of time to pay a pre-agreed price for this equipment in an emergency. This incentivizes suppliers to stockpile such equipment without fear that the government will seize it without compensation. ↩

[29] Why am I more worried about large actors for seizing power, but small actors for causing destruction? Because the dynamics are

different. Seizing power is about whether one actor can amass enough strength to overcome everyone else—thus we should worry about the most powerful actors and/or those closest to AI. Destruction, by contrast, can be wrought by those with little power if it is much harder to defend against than to cause. It is then a game of defending against the most *numerous* threats, which are likely to be smaller actors. ↩

[30] This might sound like it is in tension with my point that attack and defense may be more balanced with cyberattacks than with bioweapons, but my worry here is that if a country's AI is the most powerful in the world, then others will not be able to defend even if the technology itself has an intrinsic attack-defense balance. ↩

[31] For example, in the United States this includes the fourth amendment and the Posse Comitatus Act. ↩

[32] Also, to be clear, there are some arguments for building large datacenters in countries with varying governance structures, particularly if they are controlled by companies in democracies. Such buildouts could in principle help democracies compete better with the CCP, which is the greater threat. I also think such datacenters don't pose much risk unless they are very large. But on balance, I think caution is warranted when placing very large datacenters in countries where institutional safeguards and rule-of-law protections are less well-established. ↩

[33] This is, of course, also an argument for improving the security of the nuclear deterrent to make it more likely to be robust against powerful AI, and nuclear-armed democracies should do this. But we don't know what a powerful AI will be capable of or which defenses, if any, will work against it, so we should not assume that these measures will necessarily solve the problem. ↩

[34] There is also the risk that even if the nuclear deterrent remains effective, an attacking country might decide to call our bluff—it's unclear whether we'd be willing to use nuclear weapons to defend against a drone swarm even if the drone swarm has a substantial risk of conquering us. Drone swarms might be a new thing that is less severe than nuclear attacks but more severe than conventional attacks. Alternatively, differing assessments of the effectiveness of the nuclear deterrent in the age of AI might alter the game theory of nuclear conflict in a destabilizing manner. ↩

[35] To be clear, I would believe it is the right strategy not to sell chips to China, even if the timeline to powerful AI were substantially longer. We cannot get the Chinese "addicted" to American chips—they are determined to develop their native chip industry one way or another. It will take them many years to do so, and all we are doing by selling them chips is giving them a big boost during that time. ↩

[36] To be clear, most of what is being used in Ukraine and Taiwan today are not *fully* autonomous weapons. These are coming, but not here today. ↩

[37] Our model card for Claude Opus 4.5, our most recent model, shows that Opus performs better on a performance engineering interview frequently given at Anthropic than any interviewee in the history of the company. ↩

[38] "Writing all of the code" and "doing the task of a software engineer end to end" are very different things, because software engineers do much more than just write code, including testing, dealing with environments, files, and installation, managing cloud compute deployments, iterating on products, and much more. ↩

[39] Computers are general in a sense, but are clearly incapable on their own of the vast majority of human cognitive abilities, even as they

greatly exceed humans in a few areas (such as arithmetic). Of course, things built *on top* of computers, such as AI, are now capable of a wide range of cognitive abilities, which is what this essay is about. ↩

[40] To be clear, AI models do not have precisely the same profile of strengths and weaknesses as humans. But they are also advancing fairly uniformly along every dimension, such that having a spiky or uneven profile may not ultimately matter. ↩

[41] Though there is debate among economists about this idea. ↩

[42] La richesse personnelle est un « stock », tandis que le PIB est un « flux ». Il ne s'agit donc pas d'affirmer que Rockefeller possédait 2 % de la valeur économique des États-Unis. Cependant, il est plus difficile de mesurer la richesse totale d'une nation que le PIB, et les revenus individuels varient considérablement d'une année à l'autre ; il est donc difficile d'établir un ratio dans les mêmes unités. Le ratio entre la plus grande fortune personnelle et le PIB, bien que la comparaison ne soit pas pertinente, constitue néanmoins un indicateur tout à fait raisonnable de la concentration extrême des richesses .

[43] La valeur totale du travail dans l'économie est de 60 000 milliards de dollars par an ; 3 000 milliards de dollars par an correspondraient donc à 5 % de cette valeur. Une entreprise fournissant de la main-d'œuvre à 20 % du coût des travailleurs et détenant 25 % de parts de marché pourrait gagner ce montant, même si la demande de main-d'œuvre n'augmentait pas (ce qui serait presque certainement le cas compte tenu du coût inférieur) .

[44] Pour être clair, je ne pense pas que la productivité réelle de l'IA soit encore responsable d'une part substantielle de la croissance économique américaine. Je pense plutôt que les dépenses liées aux centres de données représentent une croissance causée par des investissements anticipés, ce qui revient à ce que le marché s'attende à

une croissance économique *future* tirée par l'IA et investisse en conséquence .

[45] Lorsque nous sommes d'accord avec l'administration, nous le disons, et nous recherchons les points de convergence où les politiques partagées sont véritablement bénéfiques pour le monde. Nous aspirons à être des médiateurs impartiaux plutôt que des partisans ou des opposants à un parti politique en particulier.

[46] Je ne pense pas que cela puisse prendre plus de quelques années : à plus long terme, ils construiront leurs propres puces .

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